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Interpretable physics-based hybrid modeling with improved BPNN and transfer learning for prediction and optimization of melt pool morphology in laser cladding

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ABSTRACT

In laser cladding, the morphology of the melt pool is a critical determinant of the geometric accuracy and metallurgical quality of the clad layer, making its prediction and optimization essential for process design and quality control. To overcome the limitations of conventional data-driven models, including low accuracy and poor interpretability under small-sample conditions, this study proposes an interpretable hybrid modeling approach that integrates a physical model with an improved back-propagation neural network (BPNN). First, a physical model combining a Rosenthal point heat source and a Gaussian surface heat source is established to derive analytical relationships between process parameters (laser power, scanning speed, and powder feed rate) and melt pool morphologies (width, depth, and height), with energy balance constraints ensuring physical consistency. Second, a hybrid Golden sine and Subtraction Average-Based Optimizer (GSABO) optimization strategy is applied to enhance the BPNN, enabling effective correction of residuals between the physical model predictions and experimental data. The SHapley Additive exPlanations (SHAP) method is further employed to interpret the contribution of each process parameter to prediction errors. Additionally, a transfer learning strategy is adopted to facilitate small-sample modeling across different materials and equipment, substantially improving the model's generalization capability. Finally, a multi-objective optimization framework based on NSGA II is constructed to inversely optimize process parameters according to target melt pool morphologies. Experimental results demonstrate that the proposed physics-improved BPNN hybrid model outperforms traditional methods in prediction accuracy, interpretability, and transferability, offering an efficient and reliable solution for intelligent modeling and optimization of the laser cladding process.

1. Introduction

Laser cladding is an advanced surface modification technology that uses a high-energy laser beam to melt and deposit metal powder onto a substrate, forming a coating with enhanced wear resistance, corrosion resistance, and mechanical strength [1–3]. Compared to conventional techniques such as thermal spraying, electroplating, and arc welding, laser cladding offers significant advantages including a low dilution rate [4], high bonding strength [5], and a dense microstructure [6]. These characteristics have made it widely applicable in industries such as aerospace, hydraulic engineering, energy equipment, and high-end manufacturing. The single-track clad layer acts as the fundamental

building unit of a laser-cladded coating, and its geometric morphology decisively influences the forming quality and metallurgical bonding of the deposited component [7]. However, the laser cladding process involves strong coupling among multiple physical fields, including optical, thermal, and fluid, leading to a melt pool that undergoes continuous and rapid melting and solidification. This dynamic behavior makes it difficult to effectively observe the morphological evolution of the clad layer [8–10]. Therefore, establishing an accurate mapping model between process parameters and the geometric morphology of the clad layer is essential for achieving intelligent control and process optimization in laser cladding.

With the advancement of intelligent manufacturing technologies,

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machine learning has been widely adopted for modeling and prediction in laser cladding. Researchers have utilized algorithms such as neural networks, support vector machines, and random forest regression to predict key process characteristics, including clad geometry, dilution rate, and hardness [11–15]. These data-driven models are capable of capturing complex nonlinear mapping relationships, overcoming the constraints of traditional regression methods and opening new pathways for the intelligent development of laser cladding. However, purely data-driven models suffer from two major limitations. First, their training relies heavily on large experimental datasets, making them susceptible to overfitting when sample sizes are small [16]. Second, their internal mechanisms lack interpretability and explicit physical meaning, which often leads them to be perceived as “black-box” models [17]. Consequently, even when such models achieve high predictive accuracy, they fail to reveal the underlying physical mechanisms through which process parameters affect melt pool behavior, thus limiting their applicability and credibility in engineering practice.

To address the interpretability issue of data-driven models, the Shapley Additive exPlanations (SHAP) method, inspired by game theory, has been proposed and applied in machine learning [18]. The SHAP method quantifies the importance of each process parameter by computing the marginal contribution of input features to the output, thereby revealing the model’s decision logic. For laser cladding, SHAP analysis can reveal both the magnitude and direction of the influence of process parameters on quality-related features, while also identifying the main sources of prediction errors, thus providing a basis for further process optimization [19–21]. However, SHAP is essentially a post hoc interpretability method that lacks inherent physical constraints, and thus cannot guarantee the physical rationality of the model’s predictions [22]. Therefore, relying solely on interpretability methods cannot fundamentally resolve the issue of physical consistency in “black-box” models.

In recent years, physics-based or physics-constrained modeling in machine learning has gradually emerged as a research frontier in the field of intelligent manufacturing. By embedding physical laws, conservation principles, or analytical models into machine learning frameworks, it is possible to significantly improve prediction accuracy and generalization performance while maintaining efficient learning capability [23]. This concept has been validated in the field of laser processing. For instance, Zhu et al. [24] developed a physics-informed neural network (PINN) framework incorporating first-principles physics, which enables accurate prediction of temperature and melt pool dynamics in metal additive manufacturing using limited sample data. Peng et al. [25] integrated the PINN model with transfer learning to construct a predictive model that infers three-dimensional temperature fields from two-dimensional temperature distributions, thereby enhancing real-time temperature monitoring and feedback control in metal additive manufacturing. Wang et al. [26] proposed a physics-constrained temporal convolutional network for accurate prediction of melt pool dimensions in laser-directed energy deposition, which effectively supports the high-quality fabrication of thin-walled components. In summary, the integration of physical mechanisms with machine learning offers a new paradigm for modeling laser processing, enabling models to retain the flexibility of data-driven approaches while achieving physical interpretability and engineering generalizability.

In summary, a hybrid prediction framework integrating a physical model with an improved BPNN is proposed. Combined with SHAP-based interpretability analysis and transfer learning strategies, the framework achieves accurate prediction and optimization of melt pool morphology in laser cladding. First, an analytical physical model combining the Rosenthal heat source model and the Gaussian heat source model is established to provide physical prior knowledge for the data-driven model. Subsequently, to address the issues of local optima and unstable convergence in traditional BPNN, a GSABO optimization strategy is introduced to perform global optimization of network weights and thresholds, effectively correcting the residuals of physical model

predictions and improving the model’s fitting accuracy and generalization capability. On this basis, the SHAP method is employed to perform interpretability analysis of feature contributions within the model, revealing the mechanisms and physical significance of how process parameters influence the prediction process. To overcome the modeling challenges caused by small-sample distribution differences across materials and equipment, this study introduces a physics-based transfer learning approach. By freezing parameters of the source model and fine-tuning via L2-SP regularization, the predictive model achieves efficient transfer and knowledge retention under different conditions. Finally, a multi-objective optimization framework based on NSGA II is developed, taking cladding efficiency, dilution rate, and aspect ratio as joint optimization objectives to achieve inverse optimization of process parameters. Through the above research, this study aims to establish an intelligent modeling and optimization system for laser cladding that integrates physical constraints, interpretability, and cross-domain generalization capability, providing a systematic theoretical framework and guidance for process parameter design and clad-layer quality control.

2. Experiment and data

2.1. Materials and equipment

In this experiment, 20Cr13 stainless steel (10 mm × 10 mm × 70 mm) was used as the substrate for laser cladding. Before the experiment, the substrate surface was polished with an angle grinder to remove the oxide layer and surface impurities, then cleaned with acetone to eliminate surface oil contamination, and finally dried for subsequent use. Iron-based 15-5PH powder with a particle size range of 53–150 μm was used as the cladding material. Prior to the experiment, the powder was placed in a 200 °C drying oven for 30 min to ensure its flowability. This experiment is designated as Experiment 1, and the main chemical compositions of the substrate and powder are listed in Table 1.

To achieve transfer learning for predicting single-track clad layer morphology in laser cladding across different materials and equipment conditions, laser cladding experiments were conducted using 2507 duplex stainless steel (10 mm × 10 mm × 70 mm) as the substrate and Stellite-6 cobalt-based powder (53–150 μm) as the cladding material. The same pretreatment procedures were performed prior to the experiment to ensure a clean substrate surface and good powder flowability. This experiment is designated as Experiment 2, and the main chemical compositions of the substrate and powder are listed in Table 2.

In this study, Experiment 1 employed an RFL-A3000D semiconductor laser as the light source, with an ABB six-axis robot controlling a coaxial powder-feeding laser cladding head. The setup was equipped with auxiliary devices including an electrical control system, a dual-hopper negative-pressure powder feeder, an AC voltage stabilizer, and a water-cooling unit. During the laser cladding process, 99.999% high-purity argon was used as the shielding gas to protect the melt pool. Experiment 2 used an RFL-C3000 fiber laser generator as the light source, with a FANUC robot controlling a TS-6.0A coaxial powder-feeding laser cladding head. The system was equipped with an electrical control system, an ECPF-DCM-D1-2-P2 powder feeder, an AC voltage stabilizer, and a water-cooling unit. During the laser cladding process, 99.999% high-purity argon was used to protect the melt pool. The laser cladding systems used in Experiments 1 and 2 are shown in Fig. 1 (a) and (b), respectively, while a schematic diagram of the cladding process is shown in Fig. 1(c).

Table 1

The main components of materials for Experiment 1 (wt.%).

Material	C	Mn	Si	Cr	Ni	Cu	Nb	Fe
20Cr13	0.18	0.60	0.36	11.22	0.27	/	/	Bal.
15-5PH	0.03	0.84	0.38	15.41	4.66	3.71	0.30	Bal.

Table 2

The main components of materials for Experiment 2 (wt.%).

Material	C	N	Si	Mo	W	Mn	Cr	Ni	Fe	Co
2507	0.03	0.28	0.75	4.18	/	1.0	24.5	6.8	Bal.	/
Stellite6	1.05	/	1.20	1.08	4.91	0.74	29.54	1.81	2.25	Bal.

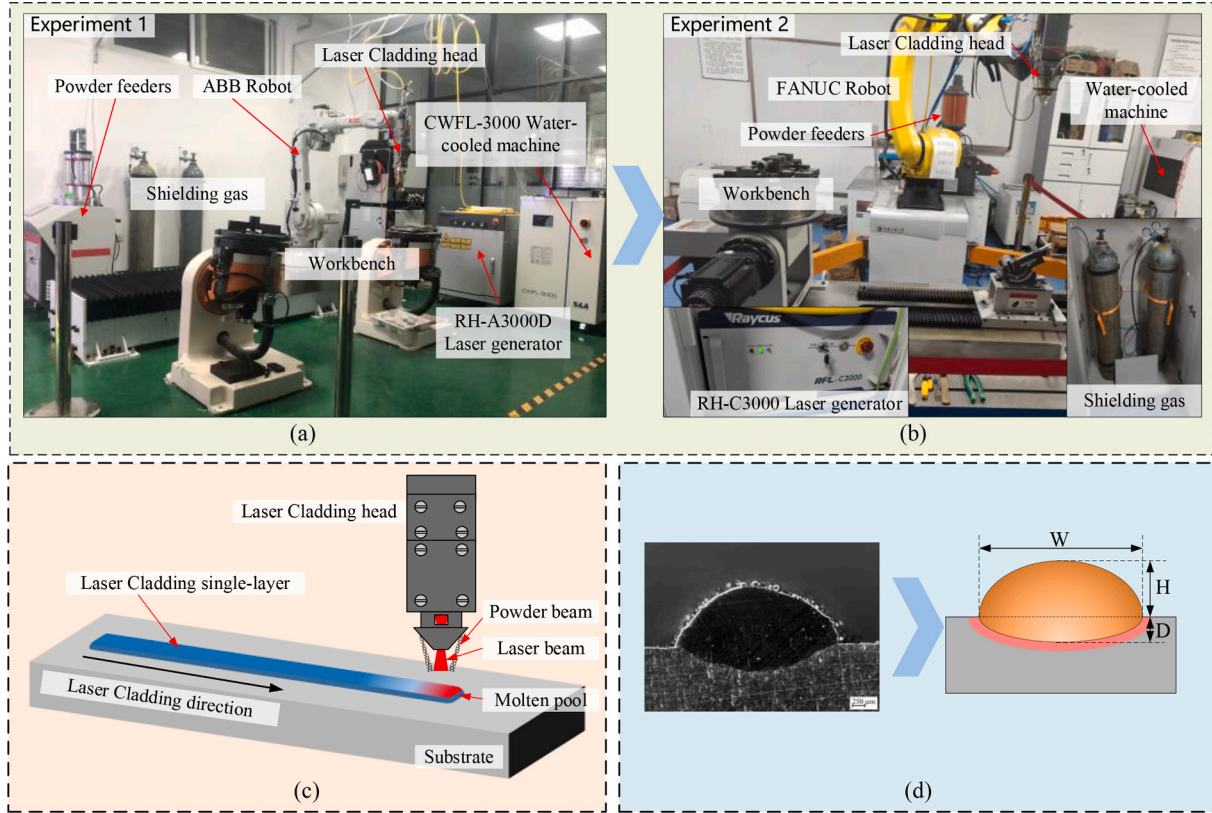


Fig. 1. Laser cladding system and process characterization. (a) Laser cladding system of Experiments 1; (b) Laser cladding system of Experiments 2; (c) schematic diagram of the cladding process; (d) morphology data acquisition.

2.2. Experimental design and data analysis

2.2.1. Laser cladding experiment for modeling

The full factorial experimental design enables a comprehensive investigation of all possible combinations of factor levels, allowing for precise estimation of both main effects and interaction effects. This approach provides a complete and reliable foundation for subsequent research. In the laser cladding process, parameters such as laser power, scanning speed, powder feed rate, spot diameter, defocus distance, and carrier gas flow rate critically influence the forming quality of the clad layer. This study focuses on the key controllable variables in engineering applications, selecting laser power (P), scanning speed (V), and powder feed rate (F) as experimental input parameters. Based on preliminary tests, their reasonable ranges were determined as 1140–1500 W for laser power, 10–14 mm/s for scanning speed, and 17.1–20.7 g/min for powder feed rate. A three-factor, five-level full factorial single-track laser cladding experiment for Experiment 1 was designed accordingly, as shown in Table 3. To eliminate the influence of irrelevant variables on the experimental results, the spot diameter was fixed at 2 mm, the defocus distance at 0 mm, and the carrier gas flow rate at 10 L/min.

2.2.2. Data analysis

Laser energy density (E_d) is a comprehensive parameter that integrates laser power (P), scanning speed (V), and spot diameter (d), and is commonly used to characterize the intensity of laser energy absorbed

Table 3

The full-factorial experimental design scheme of Experiment 1.

Process parameters	Symbol	Unit	Level				
			1	2	3	4	5
Laser power	P	W	1140	1230	1320	1410	1500
Scanning speed	V	mm/s	10	11	12	13	14
Powder feeding rate	F	g/min	17.1	18.0	18.9	19.8	20.7

by the material during laser processing. Its mathematical expression is given in Eq. (1) [27]. By appropriately controlling E_d , precise regulation of the processing conditions can be achieved, making it one of the key parameters in laser manufacturing. Therefore, analyzing the influence of E_d on melt pool morphology in laser cladding is essential for gaining deeper insight into the evolution mechanisms of the melt pool geometry. It should be noted that the surface laser energy density commonly used in powder-fed laser cladding only represents the laser input intensity per unit scanning area and does not account for the effects of powder feeding rate (F) on powder stream attenuation, energy distribution, or mass input per unit time. Therefore, to more objectively evaluate the individual effects and interaction effects of process parameters, Fig. 2(a) presents the overall correlation trend between E_d and melt pool geometry, while Fig. 2(b–d) further examine the influence of powder feed rate (F) on width (W), depth (D), and height (H) of melt pool under varying

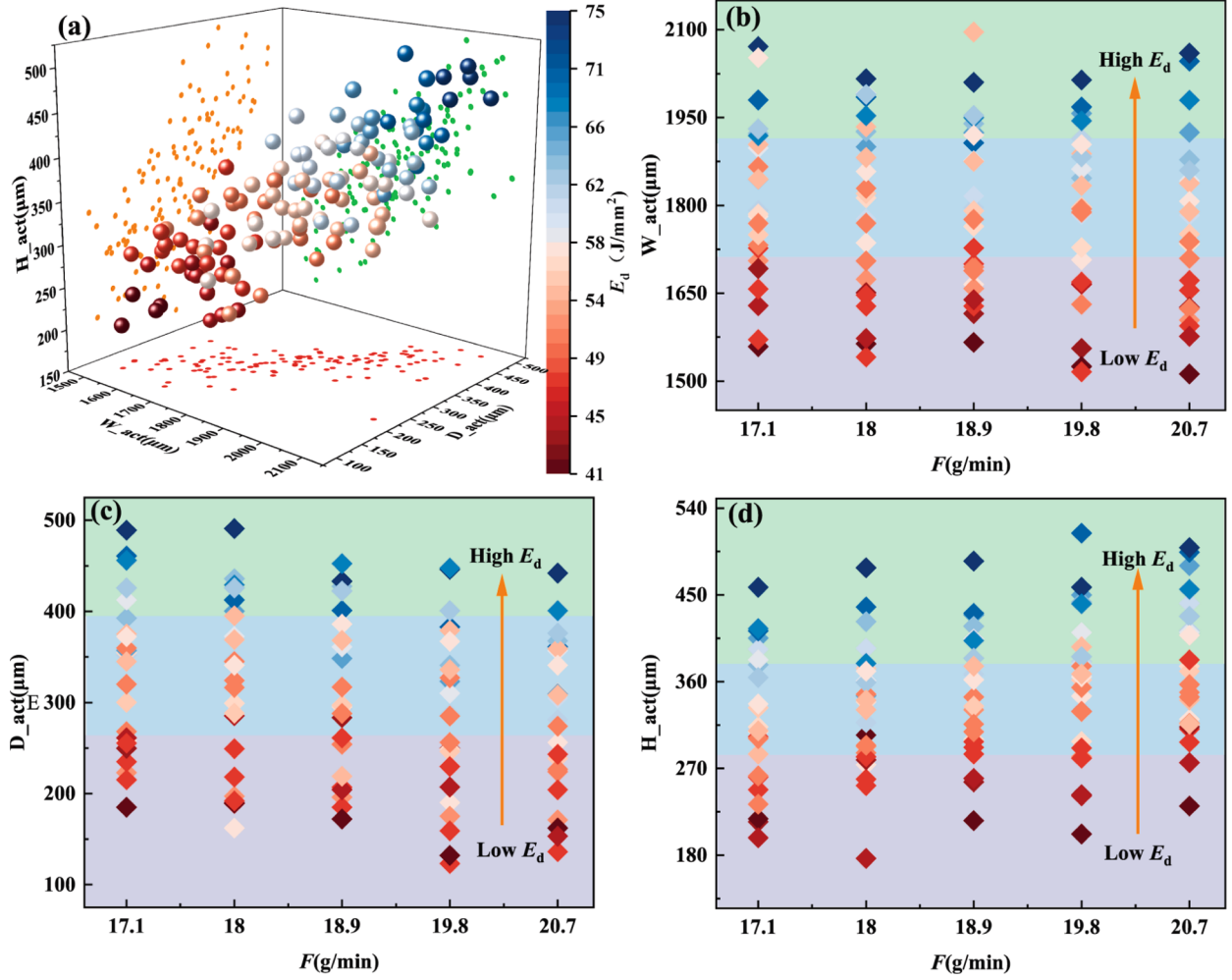


Fig. 2. Full-factor experimental dataset. (a) Effect of E_d on clad layer morphology; (b) combined effects of E_d and F on width; (c) combined effects of E_d and F on depth; (d) combined effects of E_d and F on height.

E_d conditions.

$$E_d = \frac{P}{D \cdot V} \quad (1)$$

As shown in Fig. 2(a), the W , D and H of the clad layers all exhibit an overall increasing trend with increasing E_d . This indicates that higher energy input enlarges the melting range of the substrate, increases the melt pool volume, and promotes the participation of more powder in melting and deposition. Fig. 2(b–d) further demonstrate that, under the same F , all morphological features increase with increasing E_d (as indicated by the color gradient), suggesting that a higher E_d enables more powder to participate in melting and deposition and thereby improves powder utilization. Under similar or identical E_d conditions, variations in F still lead to discernible changes in W , D , and H . However, the magnitude of these changes is weaker than the dominant effects induced by P and V . Specifically, the influence of F on W and D is relatively weak, whereas its positive effect on H is more pronounced. These observations indicate that F exerts a secondary yet non-negligible modulating effect on melt pool morphology by influencing laser energy absorption by the powder stream and the mass input per unit time.

Laser cladding is a complex process involving the coupling of multiple physical fields, including laser radiation, powder flow, and shielding gas. To accurately establish the mapping relationship between process parameters and melt pool morphology of the clad layer, it is necessary to analyze the correlations between the process parameters (P , V , F) and the melt pool geometric features (W , D , H) in order to evaluate

their degrees of association. Therefore, a Pearson correlation matrix was employed for quantitative analysis, and the results are presented in Fig. 3.

According to the results of the correlation matrix, the following conclusions can be drawn. P shows a strong positive correlation with

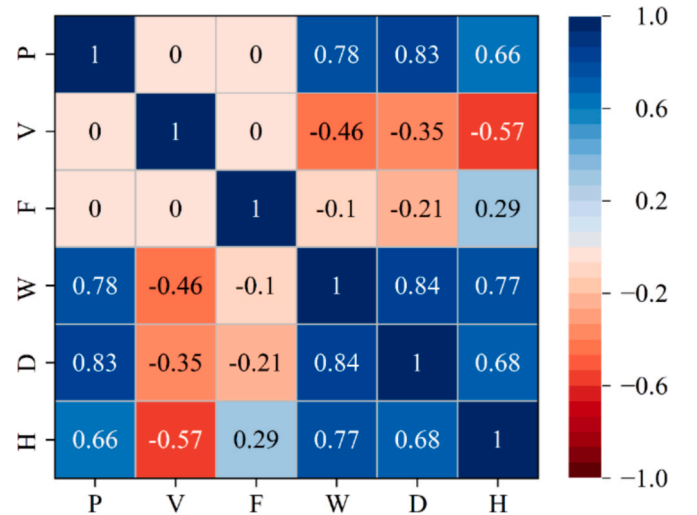


Fig. 3. Correlation matrix among the parameters.

W, D, and H of the cladding layer, with correlation coefficients of 0.78, 0.83, and 0.66, respectively. As P increases, the energy input becomes larger, the melting range of the substrate expands, W and D both increase significantly, and more powder is melted and deposited, resulting in an increase in H. V shows a negative correlation with the W, D, and H of the cladding layer, with correlation coefficients of -0.46 , -0.35 , and -0.57 , respectively. As V increases, the interaction time between the laser beam and the material becomes shorter, and the melting range of the substrate becomes narrower. Therefore, W and D decrease, and the amount of powder deposited per unit time decreases, leading to a decrease in H. F shows a weak negative correlation with W and D of the cladding layer, with correlation coefficients of -0.1 and -0.21 , and a weak positive correlation with H, with a coefficient of 0.29 . As F increases, a larger proportion of laser energy is absorbed by the powder, reducing the laser energy is absorbed by the substrate. Consequently, W and D show a slight decreasing trend, while the increased powder deposition per unit time causes H to increase. In addition, W, D, and H show strong positive correlations with each other, indicating that when one output increases, the other two also tend to increase.

In summary, the results of the correlation matrix analysis are consistent with the trend analysis based on E_d . The strong positive correlation of P and the strong negative correlation of V jointly determine the overall increasing trends of W, D, and H with increasing E_d . Meanwhile, the weak correlation between F and H, together with the weak negative correlations between F and W as well as D, further indicates that although the F is not a dominant factor, it exerts a non-negligible modulating effect on melt pool morphology.

2.2.3. Laser cladding experiment for transfer learning

Based on preliminary experiments, the reasonable parameter ranges for laser cladding Stellite6 coatings on 2507 duplex stainless steel were determined, and a three-factor, five-level orthogonal experiment was designed for Experiment 2, as shown in Table 4. To eliminate the interference of irrelevant variables on the experimental results, the spot diameter was set to 2 mm, the defocus distance to 0 mm, and the carrier gas flow rate to 10 L/min. After the experiment, the cladding layers were treated following the same procedure as described in Section 2.2.1 for morphology observation and data collection.

3. Methodology

3.1. Physical prediction model

In laser cladding, the morphology of the melt pool directly determines the geometric accuracy and forming quality of the clad layer. However, this process involves complex multi-physics coupling, and the melt pool undergoes continuous and rapid melting and solidification, making it difficult to observe or control directly through experimental means. The morphology of a single-track clad layer serves as an intuitive and measurable external manifestation of both the melt pool morphology and its dynamic behavior [28]. Therefore, by developing a reasonable physical model to establish the mapping relationship between process parameters and clad layer morphology, theoretical support can be provided for process optimization and control.

To address this, this study combines the Rosenthal moving point heat source model with the Gaussian surface heat source model to reveal the

mapping relationship between process parameters (P , V , F) and the clad layer morphology (W, D, H) [29,30]. The Rosenthal model describes the fundamental laws of heat conduction; however, due to its point heat source assumption, it cannot accurately represent the spatial distribution of laser beam energy. In contrast, the Gaussian heat source model more accurately reflects the actual distribution characteristics of laser energy density. The combination of the two models retains the analytical advantages of Rosenthal's formulation while yielding a temperature field closer to reality, thereby improving the prediction accuracy of clad layer morphology. The schematic diagram of the heat source model is shown in Fig. 4.

In the modeling process, the following assumptions are made to simplify the analysis:

- (1) The material is isotropic, and its thermal properties are taken as constants at room temperature.
- (2) The laser heat source is approximated as an ideal point heat source, with energy distribution errors corrected using a Gaussian function.
- (3) Effects such as convection, thermal radiation, mass transfer, and Marangoni flow within the melt pool are neglected.
- (4) The cross-section of the cladding layer is approximated as a parabolic shape.
- (5) The melt pool boundary is defined by the isothermal surface at the material's melting point ($T = T_m$).

Based on the above assumptions, analytical expressions of W, D, and H with respect to the process parameters are derived separately. The equation for W primarily captures the relationship between lateral heat conduction and energy distribution. The equation for D further accounts for the influence of the powder-feeding process on energy allocation. The equation for H is derived from mass conservation of powder per unit time, incorporating corrections for powder utilization and energy density to reflect the contribution of deposition efficiency to layer height. Finally, by introducing an energy-conservation constraint to validate the results, the model is ensured to be not only formally self-consistent but also physically consistent with the fundamental laws governing the cladding process. At the macroscopic level, the proposed physical prediction model elucidates the mapping between process parameters (P , V , F) and melt pool morphology (W, D, H), thereby providing interpretable theoretical support for subsequent machine-learning-based hybrid frameworks.

3.1.1. Heat source model

The determination of melt pool temperature is crucial for constructing predictive models of the laser cladding process. To address the high cost and heavy computational burden associated with online measurement and numerical simulation, this study performs a theoretical analysis of the melt pool temperature using equations based on the Rosenthal heat source model. For a moving point heat source on a semi-infinite body, Rosenthal derived an analytical solution for the quasi-steady-state temperature field. By incorporating the Rosenthal moving heat source formulation with laser energy absorption corrections, the analytical expression of the quasi-steady-state temperature field is given as follows:

$$\left\{ \begin{array}{l} T(x', y, z) = T_0 - \frac{nP_w}{2\pi kR} \exp\left[-\frac{v(R+x')}{2\alpha}\right] \\ x' = x - vt \\ R = \sqrt{x'^2 + y^2 + z^2} \\ \alpha = k/\rho C_p \end{array} \right. \quad (2)$$

where T is the instantaneous temperature, T_0 is the initial temperature, n is the temperature calibration coefficient, P_w is the effective laser power absorbed by the substrate and powder, k is the thermal conductivity, R is

Table 4
Orthogonal experimental design for Experiment 2.

Process parameters	Symbol	Unit	Level				
			1	2	3	4	5
Laser power	P	W	1600	1700	1800	1900	2000
Scanning speed	V	mm/s	7	8	9	10	11
Powder feeding rate	F	g/min	7.2	9.6	12	14.4	16.8

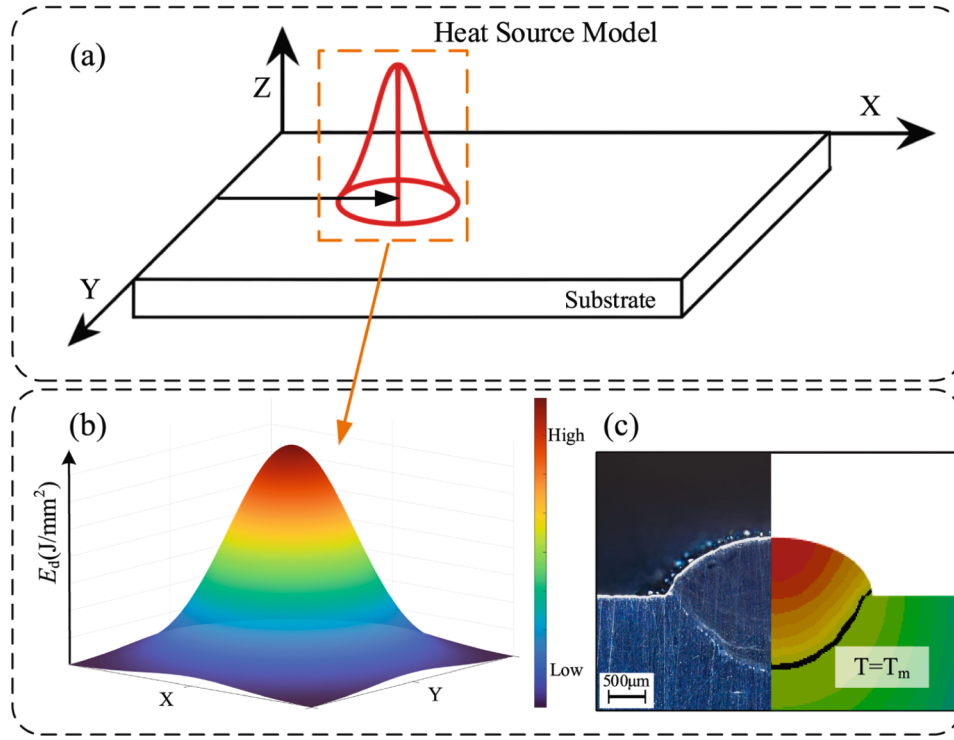


Fig. 4. Schematic diagram of the heat source model. (a) Schematic diagram of the heat source in laser cladding; (b) 3D thermal map; (c) Schematic diagram of the melt pool morphology.

the distance between the interest point and the melt pool center, α is the thermal diffusivity, ρ is the metal powder density, and C_p is the specific heat capacity.

The value of P_w in Eq. (2) is calculated according to the following equation:

$$\begin{cases} P_w = \beta_a P_{at} + \eta_p (P - P_{at}) \\ \beta_a = \beta_w [1 + a_w \tan^{-1}(2h_0/r_b)] \\ P_{at} = P [\exp(-(k_1 F + k_2 F^2))] \end{cases} \quad (3)$$

where P is the nominal input laser power, β_a is the substrate absorptivity considering the Brewster effect and the effective laser incidence angle [26], a_w is the Brewster coefficient, h_0 is the laser incident height, r_b is the laser spot radius, P_w denotes the absorbed power of the substrate, and η is the powder utilization rate. P_{at} represents the laser power after attenuation by the powder stream, and its exponential decay form originates from the Beer-Lambert law, which describes the propagation of radiation through an absorbing medium [31]. Considering that the “effective optical thickness” of the powder stream in powder-fed laser cladding exhibits a nonlinear dependence on the powder feed rate F , the attenuation exponent is approximated as a second-order function of F within the investigated range. The k_1 and k_2 are calibration coefficients characterizing laser attenuation through the powder stream, which are identified by inverse fitting based on melt pool width data. The detailed determination procedure is provided in Appendix A.

The laser heat source distribution is approximated as a Gaussian heat source, where the power decays radially along the spot diameter. The Gaussian surface heat source formula is used to correct the “point heat source approximation error” in the Rosenthal equation. The power density reaches its maximum at the heat source center ($x' = 0, y = 0$) and minimum at the edges, ensuring that the melt pool temperature is highest at the center and decreases to T_m at the boundary.

$$q''(x', y) = \frac{2P_w}{\pi r_b^2} \exp\left(-\frac{2(x'^2 + y^2)}{r_b^2}\right) \quad (4)$$

3.1.2. Model derivation process

The melt pool boundary is defined by the isothermal surface at the melting point: $T = T_m$. To determine the maximum melt width, the case directly beneath the heat source ($x = 0, z = 0$) is considered. In this case, $R = |y|$, and the temperature distribution is simplified as follows:

$$T(y) - T_0 = \frac{nP_w}{2\pi k|y|} \exp\left[-\frac{v(|y| + 0)}{2\alpha}\right] \quad (5)$$

At the melt pool boundary, $T(y) = T_m$, $|y| = W/2$. The equation becomes:

$$T_m - T_0 = \frac{nP_w}{\pi k W} \exp\left[-\frac{vW}{4\alpha}\right] \quad (6)$$

Rearranging the Eq. (6) and simplifying it into an iterative form:

$$W = \frac{nP_w}{\pi k (T_m - T_0)} \frac{1}{\sqrt{\frac{vW}{4\alpha} + 1}} \quad (7)$$

This equation reflects the balance between lateral energy input and heat conduction, as P increases or V decreases, the W correspondingly increases.

When ($x = 0, y = 0$), the temperature distribution depends only on the depth direction z :

$$T(z) - T_0 = \frac{nP_w}{2\pi k|z|} \exp\left[-\frac{v(|z| + 0)}{2\alpha}\right] \quad (8)$$

Rearranging Eq. (8) and simplifying it into an iterative form, as shown in Eq. (9):

$$D = \frac{nP_w}{2\pi k (T_m - T_0)} \frac{1}{\sqrt{\frac{vD}{2\alpha} + 1}} \quad (9)$$

As indicated by Eq. (9), the physical model for melt pool depth is explicitly dependent only on laser power and scanning speed. According to the data analysis presented in Section 2.2.2, the F still exhibits a weak negative correlation with D , with a correlation coefficient of -0.21 . If

the effect of F is neglected, the discrepancy between the predicted and experimental values of D will progressively increase as F increases. During powder-feed laser cladding, when F is relatively low, a smaller portion of laser energy is allocated to the powder stream while more energy is delivered to the substrate, resulting in an increase in melt pool depth. Conversely, an increase in F leads to a reduction in melt pool depth. Therefore, analogous to the influence of F on laser attenuation described by the Beer-Lambert law, the effect of F on D is inherently nonlinear. To account for this effect, a correction factor $f(F)$ related to the powder feed rate F is introduced to modify the influence of the nominal energy partition assumed in the analytical solution, yielding the following prediction formula for melt pool depth:

$$\begin{cases} D = \frac{nP_w}{2\pi ek(T_m - T_0)} \cdot \frac{1}{\sqrt{\frac{VD}{2\alpha}} + 1} \cdot f(F) \\ f(F) = \exp(-k_3F - k_4F^2) \end{cases} \quad (10)$$

where k_3 and k_4 are calibration coefficients describing the influence of the powder feed rate F on D , which are identified through inverse fitting based on experimental depth data. The detailed procedure and results are provided in [Appendix A](#). This formulation indicates that D depends not only on the overall energy input but also on the competitive absorption of laser energy between the powder stream and the substrate.

The H is related to the powder deposition mass per unit time and the cross-sectional geometry. The effective mass of powder deposited per unit time is given by Eq. (11):

$$M_{\text{dep}} = \eta_p \frac{F}{60} \quad (11)$$

where η_p is the powder utilization rate.

The cross-sectional area of the clad layer is approximated as a parabolic shape and can be obtained using the parabolic area formula as follows:

$$\begin{cases} h(y) = H \left(1 - \left(\frac{2y}{W} \right)^2 \right), y \in [-W/2, W/2] \\ A_{\text{clad}} = \int_{-W/2}^{W/2} h(y) dy = H \int_{-W/2}^{W/2} \left(1 - \frac{4y^2}{W^2} \right) dy = \frac{2}{3} WH \end{cases} \quad (12)$$

where $h(y)$ denotes the variation of the clad layer height along the y -direction, and A_{clad} represents the approximated cross-sectional area of the clad layer.

The mass of the cladding layer formed per unit time is:

$$M_{\text{clad}} = A_{\text{clad}} \cdot v \cdot \rho = \frac{2}{3} WHv\rho \quad (13)$$

Based on mass conservation $M_{\text{dep}} = M_{\text{clad}}$, the equation for H is obtained as:

$$H = \frac{\eta_p F}{40\rho vW} \quad (14)$$

As indicated by Eq. (14), the clad layer height H is explicitly related to the F and the V . According to the data analysis in [Section 2.2.2](#), when the powder feed rate F is kept constant, an increase in the input energy density E_d leads to an increase in the clad layer height, indicating that more powder participates in the melting and deposition process and that the effective powder utilization is enhanced. By introducing a correction factor related to laser power P into the height prediction formula in Eq. (14), the systematic deviation caused by the assumption of a nominal powder utilization rate can be effectively corrected. To avoid sensitivity of parameter calibration to the absolute power scale and to ensure that the correction term remains invariant at a reference condition, normalization is performed at the mean laser power P_0 , and a semi-empirical correction function with respect to P is defined as $f(P)$.

Thereby yielding the corrected prediction formula for the H as follows:

$$\begin{cases} H = \frac{\eta_p F}{40\rho vW} \exp \left[\beta_H \left(\frac{P}{P_0} - 1 \right) \right] \cdot f(P) \\ f(P) = \exp \left[\beta_H \left(\frac{P}{P_0} - 1 \right) \right] \end{cases} \quad (15)$$

where β_H is the calibration coefficient for the clad layer height, and its specific value and identification procedure are provided in [Appendix A](#). As the F increases, the H increases accordingly; however, it is also influenced by the coupled effects of the V and the P .

3.1.3. Energy constraints

In the cladding process, the input laser energy must satisfy the energy required to melt the powder and the substrate; otherwise, the predictions lose physical meaning. Therefore, an energy conservation constraint is introduced to calibrate the model results. Energy required to melt a unit mass of powder:

$$E_{\text{pow}} = C_{p,\text{pow}}(T_m - T_0) + L_{\text{pow}} \quad (16)$$

where $C_{p,\text{pow}}$ is the specific heat capacity of the powder; L_{pow} is the latent heat of solid-liquid phase transformation of the powder. Total heat required to melt the powder per unit time:

$$Q_{\text{pow}} = M_{\text{pow}} \cdot [C_{p,\text{pow}}(T_m - T_0) + L_{\text{pow}}] \quad (17)$$

Energy required to melt a unit mass of the substrate:

$$E_{\text{sub}} = C_{p,\text{sub}}(T_m - T_0) + L_{\text{sub}} \quad (18)$$

where $C_{p,\text{sub}}$ is the specific heat capacity of the substrate; L_{sub} is the latent heat of the substrate's solid-liquid phase transition. Total heat required to melt the substrate per unit time:

$$Q_{\text{sub}} = M_{\text{sub}} \cdot [C_{p,\text{sub}}(T_m - T_0) + L_{\text{sub}}] \quad (19)$$

According to the law of energy conservation, the total heat constraint per unit time is:

$$M_{\text{pow}} \cdot [C_{p,\text{pow}}(T_m - T_0) + L_{\text{pow}}] + M_{\text{sub}} \cdot [C_{p,\text{sub}}(T_m - T_0) + L_{\text{sub}}] < P_w \cdot \Delta t \quad (20)$$

This constraint ensures that the predicted W , D , and H are energetically self-consistent, meaning that the input energy is sufficient to sustain the melting process. If the model's predictions exceed the bounds imposed by energy conservation, it indicates that the assumptions or parameter settings require revision. By imposing this energy constraint, the physical model's rationality is ensured, providing a solid foundation for subsequent integration with machine learning models.

3.2. Improved BPNN prediction model

3.2.1. Backpropagation neural network

During the laser cladding process, significant coupling effects exist among process parameters, leading to nonlinear relationships between the process parameters and the geometric features of the clad layer. The backpropagation neural network (BPNN), owing to its strong nonlinear mapping capability and flexible network architecture, has been widely applied to multi-input-multi-output regression problems and is well suited for modeling the complex mapping relationships between laser cladding process parameters and clad layer morphology [32,33]. In this study, P , V , and F are selected as input variables, while W , D , and H are used as output variables to construct a BPNN prediction model. A schematic diagram of the network architecture is shown in [Fig. 5](#).

The workflow of the BPNN prediction model is as follows:

Step 1: Data preprocessing. To eliminate the influence of different physical dimensions on model training and to improve network convergence efficiency, the dataset is normalized as shown in Eq. (21).

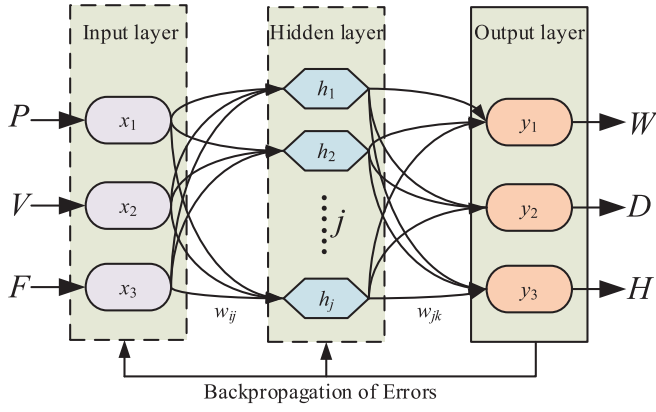


Fig. 5. Schematic diagram of the BPNN mapping model.

$$v_i^{std} = \frac{2(v_i - v_{imin})}{(v_{imax} - v_{imin})} \quad (21)$$

where v_i^{std} denotes the normalized data, v_i is the i -th original data point, and v_{imax} and v_{imin} are the maximum and minimum values of the row to which v_i belongs, respectively. Meanwhile, the output dataset of the prediction model is de-normalized using Eq. (22).

$$v_i = \frac{v_i^{std} + 1}{2} \times (v_{imax} - v_{imin}) + v_{imin} \quad (22)$$

Step 2: Network architecture determination. The BPNN consists of an input layer, a hidden layer, and an output layer. The numbers of neurons in both the input and output layers are set to three. The number of neurons in the hidden layer is preliminarily determined using the empirical formula in Eq. (23) and further adjusted within a certain range to obtain an optimal network architecture.

$$p = \sqrt{m + n} + a \quad (23)$$

where m is the number of neurons in the input layer, p is the number of neurons in the hidden layer, n is the number of neurons in the output layer, and a is typically chosen as an integer between 1 and 10.

Step 3: Model training and parameter updating. After initializing the network weights and thresholds, the network output is computed through forward propagation, and the backpropagation algorithm is employed to minimize the error between the predicted values and the ground truth. The Sigmoid function is adopted as the activation function of the hidden layer, as expressed in Eq. (24). The training process is terminated when the target error or the maximum number of iterations is reached.

$$f(x) = \frac{1}{1 + e^{-x}} \quad (24)$$

3.2.2. GSABO strategy for optimizing BPNN model

Although the BPNN algorithm offers strong nonlinear mapping capability, it tends to fall into local optima and requires large training datasets. Moreover, the choice of initial parameters can significantly affect the accuracy of the resulting predictive model. To improve the accuracy of the BPNN model, appropriate initial parameters must be selected.

The Subtraction Average-Based Optimizer (SABO), proposed by Pavel Trojovský and colleagues, is a mathematics-driven optimization algorithm [34]. It updates the positions of population members in the search space using the subtraction average of individuals, offering strong search capability and fast convergence. SABO can be used to optimize the weights and thresholds of a BPNN, yielding a more accurate input-output mapping and improving the model's prediction accuracy [35]. Because SABO does not use the global best at each iteration but

instead updates via the subtraction average of all particle positions, it can easily fall into local optima when particle initialization is poor. Therefore, when particle fitness does not change in the current iteration, the particle positions need to be updated.

Inspired by the sine function, the Golden Sine (GS) algorithm performs iterative optimization using the mathematical sine function, exhibits strong global search capability. The algorithm introduces the golden ratio coefficient during position updates to thoroughly explore local regions, thereby achieving a good balance between global exploration and local exploitation [36]. In summary, updating the weights and thresholds of the BPNN with the hybrid Golden sine and Subtraction Average-Based Optimizer (GSABO) can effectively address the tendency to get trapped in local optima. The GSABO-BPNN algorithm can be used to establish a mapping model between single-track melt pool morphology and process parameters, and its workflow is shown in Fig. 6.

The specific steps for optimizing the BPNN model with GSABO are as follows:

Step 1: Initialization. Initialize the population using a chaotic mapping method to ensure particle diversity and accelerate convergence.

Step 2: Selection of the fitness function and fitness evaluation. Specifically, for any candidate solution, the network is first evaluated through forward propagation on the training dataset, and the fitness function is defined as the mean squared error (MSE) of the output variables, as expressed in the following Eq. (25):

$$F_i = \frac{1}{N} \sum_{i=1}^N \|\hat{y}_i(\mathbf{X}_i; \boldsymbol{\theta}) - \mathbf{y}_i\|^2 \quad (25)$$

where $\mathbf{X}_i$ denotes the input sample matrix of process parameters, $\mathbf{y}_i$ represents the output sample matrix of clad layer morphology, $\boldsymbol{\theta} = [\mathbf{w}_1, \mathbf{b}_1, \mathbf{w}_2, \mathbf{b}_2]$ is the parameter vector of the BPNN searched by the GSABO algorithm, $\hat{y}_i$ is the predicted output of the BPNN under the parameter vector $\boldsymbol{\theta}$, and N is the number of samples.

Step 3: At the beginning of each iteration, search and update particle positions according to the SABO Eq. (26).

$$A - vB = \text{sign}(F(A) - F(B))(A - \vec{v} * B) \quad (26)$$

where, “ $-v$ ” is a special operation proposed in SABO, referred to as the v -subtraction between search agent B and search agent A . The population matrix E is searched and updated according to Eqs. (27) and (28):

$$E_i^{new} = E_i + \vec{r} * \frac{1}{N} \sum_{j=1}^N (E_i - vE_j) \quad (27)$$

$$E_i = \begin{cases} E_i^{new}, & F_i^{new} < F_i; \\ E_i, & \text{else} \end{cases} \quad (28)$$

where E_i^{new} and E_i represent the latest iteration position and the previous generation iteration position of the i -th individual, and F_i^{new} and F_i represent the latest fitness value and the previous generation's fitness value, respectively.

Step 4: If SABO fails to reduce the fitness, apply the GS algorithm to search and update the particle positions, as shown in Eqs. (28) and (29).

$$\begin{cases} X_1 = a * (1 - t) + b * t \\ X_2 = a * t + b * (1 - t) \end{cases} \quad (29)$$

$$E_i^{new} = E_i |\sin(r_1)| - r_2 \sin(r_1) |X_1 E_{ibest} - X_2 E_i| \quad (30)$$

where X_1 and X_2 are golden ratio coefficients; a and b are the initial values for the golden-ratio search; t denotes the golden-ratio proportion; E_i^{new} is the latest iterative position of the i -th individual; E_i and E_{ibest} are

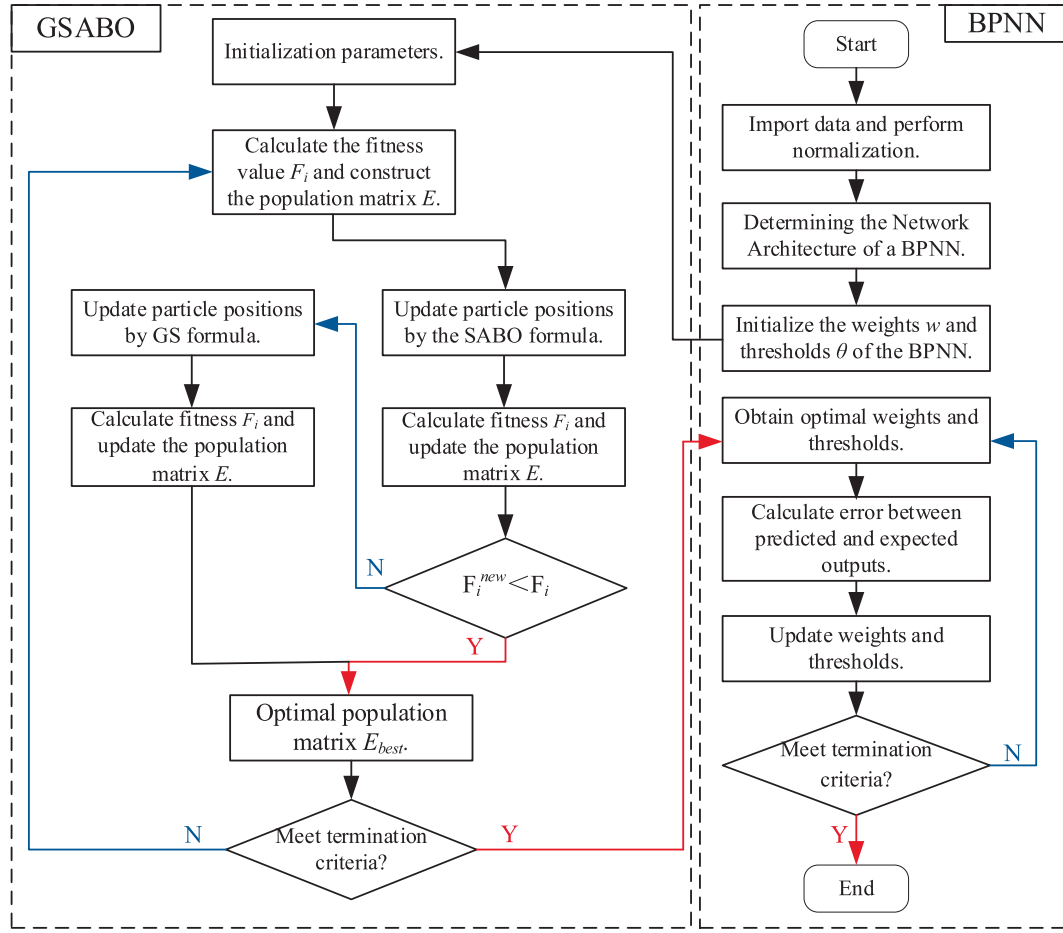


Fig. 6. Flow chart of GSABO-BPNN.

the previous and the best positions of the i -th individual, respectively.

Step 5: Save the optimal population matrix E_{best} .

Step 6: Determine whether the termination condition is satisfied. If satisfied, terminate the program and output the optimal combination of weights and thresholds for the BPNN model; otherwise, repeat Steps 2–5. The GSABO algorithm iteratively updates the positions of population individuals to continuously reduce the fitness function, ultimately yielding a globally optimal set of initial network parameters. After the global optimization by GSABO, the obtained optimal weights and thresholds are used as the initial parameters of the BPNN, followed by local gradient-based fine-tuning using back-propagation. In this manner, the global search capability of GSABO is combined with the local convergence efficiency of the BPNN, thereby improving the prediction accuracy and stability of the model.

3.3. Shapley additive explanation method

Although data-driven machine learning models exhibit high predictive accuracy in complex nonlinear problems, their internal decision-making mechanisms are often difficult to interpret intuitively, which limits their credibility in engineering applications. To enhance model interpretability, this study introduces the SHapley Additive exPlanations (SHAP) method based on cooperative game theory to quantitatively interpret the model prediction results. The SHAP method evaluates the influence of an individual feature on the model output by computing its marginal contribution across different feature combinations. Let the set of input features be denoted as N ; the SHAP value of feature i is defined as shown in Eq.(31). For any given input sample, the model prediction

can be expressed as the sum of a baseline value and the contributions of individual features, as shown in Eq.(32), where the baseline value is typically taken as the mean of the model outputs over the training dataset.

$$\varphi_i = \sum_{S \subseteq N \setminus \{i\}} \frac{|S|!(|N| - |S| - 1)!}{|N|!} [f(S \cup \{i\}) - f(S)] \quad (31)$$

$$f(x) = \varphi_0 + \sum_{i=1}^N \varphi_i \quad (32)$$

where φ_0 is typically set to the mean of the output variable over the training set as the baseline. This decomposition satisfies the principles of additivity and consistency, facilitating unified interpretation at both global and local scales.

For data-driven models, SHAP analysis is mainly employed to reveal the relative importance and directional effects of the process parameters on the geometric characteristics of the clad layer, thereby elucidating the mechanisms by which different process parameters influence forming quality. For the physics-machine learning hybrid models, SHAP analysis is further used to interpret the contributions of process parameters to the prediction residuals of the physical model. If a particular feature exhibits a large SHAP value in the residual model, it indicates that the physical model tends to overestimate or underestimate the target variable under that specific feature condition. Through the above analysis, the SHAP method not only enhances the interpretability of model predictions but also enables the identification of systematic error sources in the physical model, thereby providing quantitative guidance for its improvement and refinement.

3.4. Transfer learning

In cross-material and cross-equipment laser cladding modeling, the source and target domains often have non-overlapping ranges of process parameters. Directly performing transfer learning tends to induce feature distribution shift, thereby diminishing prediction accuracy. Therefore, within a physics–data hybrid framework, this study develops a physics-based transfer learning pipeline to enhance cross-domain adaptability and generalization performance. The workflow is shown in Fig. 7.

Specifically, physical models are employed to uniformly adjust process parameters across source and target domains, mapping input variables from different experimental conditions into a comparable physical parameter space. This step effectively mitigates the adverse effects of distribution mismatch on transfer learning, enabling knowledge from the source domain to transfer appropriately to the target domain. On this basis, the physics-GSABO-BPNN hybrid model trained on the source domain is used as the starting point for transfer. During the transfer process, the lower-layer network parameters of the model are retained and frozen to ensure its ability to represent general regularities such as

energy input, heat conduction, and melt pool geometry [37,38]. Meanwhile, selected higher-layer parameters are unfrozen and updated so that the model can adapt to differences in material properties and equipment conditions in the target domain [39].

To effectively balance source domain knowledge retention with target domain adaptability under small-sample conditions, this study employs an L2-SP regularization strategy. This approach jointly optimizes the target domain prediction error and the source domain parameter preservation term during model fine-tuning [40]. Specifically, the loss function for transfer learning is designed as follows:

$$\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{target}} + \lambda \mathcal{L}_{\text{SP}} \quad (33)$$

where λ is the regularization coefficient, used to control the retention level of source domain knowledge; $\mathcal{L}_{\text{target}}$ is the prediction error in the target domain, represented by the mean squared error (MSE):

$$\mathcal{L}_{\text{target}} = \frac{1}{N_t} \sum_{i=1}^{N_t} \|y_i - \hat{y}_i\|^2 \quad (34)$$

$\mathcal{L}_{\text{SP}}$ is the L2-SP regularization term, used to constrain the difference

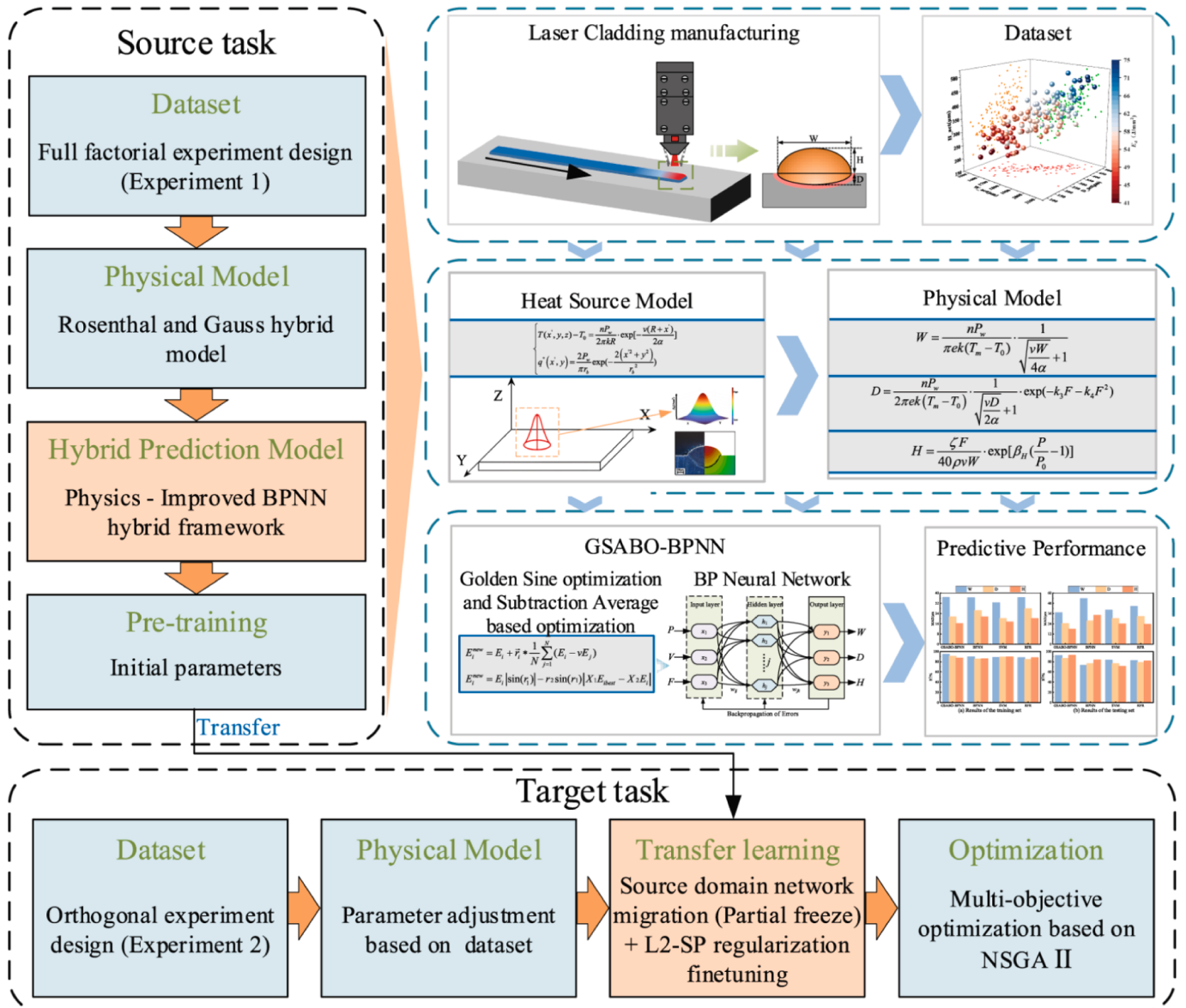


Fig. 7. Transfer Learning flow based on physical-machine learning hybrid –model for predicting and optimizing cladding layer morphology.

between the target domain model parameters θ_t and the source domain pre-trained parameters θ_s :

$$\mathcal{L}_{SP} = \|\theta_t - \theta_s\|_2^2 \quad (35)$$

Through the flow of “physical model parameter correction + source-network transfer (partial freeze) + L2-SP-regularized fine-tuning”, high-accuracy predictions are ultimately achieved under small-sample conditions in the target domain. This method not only overcomes the transfer bottleneck arising from non-overlapping process-parameter ranges in cross-material and cross-equipment laser cladding, but also markedly reduces the target domain’s reliance on large-scale experimental data, thereby providing an efficient and interpretable solution for cross-domain modeling and wider deployment of laser cladding processes.

3.5. Multi-objective optimization of NSGA II

After constructing the physics-machine learning hybrid model, performing interpretability analysis by SHAP, and enhancing generalization by cross-domain transfer learning, this study further applies the resulting model to inverse optimization of process parameters. The optimization model aims to achieve precise control of the clad-layer geometric features by appropriately tuning the combination of process parameters, while balancing energy input and process stability. Therefore, this study constructs a multi-objective optimization model based on NSGA II. The decision variables are the process-parameter vector X , expressed as:

$$X = [P, V, F]^T \quad (36)$$

The constraints of the optimization model are the permissible ranges of the process parameter levels during laser cladding, as determined by the experimental design. Therefore, the constraints on the process parameters are given by the following Eq.(37):

$$\text{s.t.} \begin{cases} 1600W \leq P \leq 2000W \\ 7\text{mm/s} \leq V \leq 11\text{mm/s} \\ 7.2\text{g/min} \leq F \leq 16.8\text{g/min} \end{cases} \quad (37)$$

Since the optimization targets in this study are the melt pool morphology indices W , D , and H , the formulated optimization model is a multi-objective minimization problem. For the application scenario of laser cladding a Stellite-6 cobalt-based coating on 2507 duplex stainless steel in Experiment 2, the industrial design specifications are explicit: W should be at least 3000 μm to ensure cladding efficiency, the dilution rate should be controlled around 0.3, and the aspect ratio should be close to 3, so as to guarantee sound metallurgical bonding and reasonable geometric characteristics of the clad layer. According to the definitions of dilution rate and aspect ratio, when D is near 430 μm and H is near 1000 μm , the morphology of clad layer is more desirable, achieving a better balance among efficiency, performance, and forming stability [13,41]. Therefore, the objective functions for optimization are as follows:

$$\text{objective function.} \begin{cases} \min |f_W(P, V, F) - 3000| \\ \min |f_D(P, V, F) - 430| \\ \min |f_H(P, V, F) - 1000| \end{cases} \quad (38)$$

The optimization model is solved by the NSGA II algorithm. The population size is set to 100, and the maximum number of evolutionary iterations is 500. The crossover operator uses a crossover probability of 0.9 and a distribution index of 15. The mutation operator adopts a mutation probability of $1/n$ (where n is the dimensionality of the decision variables, set to 3 here) and a distribution index of 20. The method is implemented in MATLAB 2025.

4. Results and discussion

4.1. Analysis of the physical model

The comparison between the physical model’s predictions and the measured values is shown in Fig. 8. As observed in the figure, the predicted results and the measured values exhibit good consistency in their trends with respect to variations in the process parameters. As P increases, W , D , and H show a pronounced upward trend, which is fully consistent with the strong positive correlations between P and W , D , and H revealed in Section 2.2. As V increases, W , D , and H tend to decrease, further corroborating the conclusion in Section 2.2 regarding the negative correlations between V and W , D , and H . When F varies, the trends in W , D , and H are weak, indicating weak correlations among them. These results demonstrate that the physical model can, to a certain extent, capture the mapping between the process parameters and the melt pool morphology.

However, the material’s thermal properties were assumed constant and practical effects such as convection and thermal radiation were neglected when constructing the physical model, which inevitably introduces discrepancies between the model predictions and the measurements (as shown in Fig. 9). In essence, during actual laser cladding process, thermal properties such as thermal conductivity and specific heat vary nonlinearly with increasing temperature. Moreover, convective redistribution of heat within the melt pool and energy losses due to thermal radiation exert non-negligible effects on the evolution of the melt pool morphology [42]. These disparities between real operating conditions and model assumptions indicate that there remains room to improve the model’s accuracy in practical engineering applications. To correct the systematic errors introduced by these assumptions, machine learning is used to learn the behavior of the physical model’s prediction residuals. This leads to the construction of a “physics-machine learning” hybrid model, which ultimately enables high-accuracy prediction of the melt pool morphology.

4.2. Comparison and evaluation

4.2.1. Comparison models and evaluation metrics.

To comprehensively validate the effectiveness of the proposed GSABO-BPNN model in predicting the morphology of laser-cladded layers, three commonly used machine learning methods were selected as comparison models, namely the backpropagation neural network (BPNN), support vector machine (SVM), and random forest regression (RFR). These models are representative baseline approaches for nonlinear regression problems and provide a reasonable reference for assessing the performance improvements achieved by the proposed model [43–45].

During the comparison, to ensure fairness among different models, all methods employ the same dataset partitioning strategy and data preprocessing procedure. Specifically, the input data are normalized to eliminate the influence of different physical scales, and the datasets are divided into training and testing sets using a fixed ratio of 4:1. Accordingly, the source-domain dataset (Experiment 1) is split at a ratio of 100:25, while the target-domain dataset (Experiment 2) is split at a ratio of 20:5. The testing set is used exclusively for final performance evaluation and comparison and does not participate in any stage of the training process. The detailed parameter settings of the comparison methods are summarized in the corresponding Table 5. Both the GSABO-BPNN and BPNN models are trained for 500 iterations to obtain the final trained networks. For the neural network models, the Adam algorithm is adopted as the optimizer for network training, and the stopping criteria are determined by the built-in convergence rules of the algorithm [46]. During training, the mean squared error (MSE) is used as the loss function to quantify the discrepancy between the predicted values and the measured values. This loss function guides the training gradients through backpropagation, thereby enabling the gradual optimization of

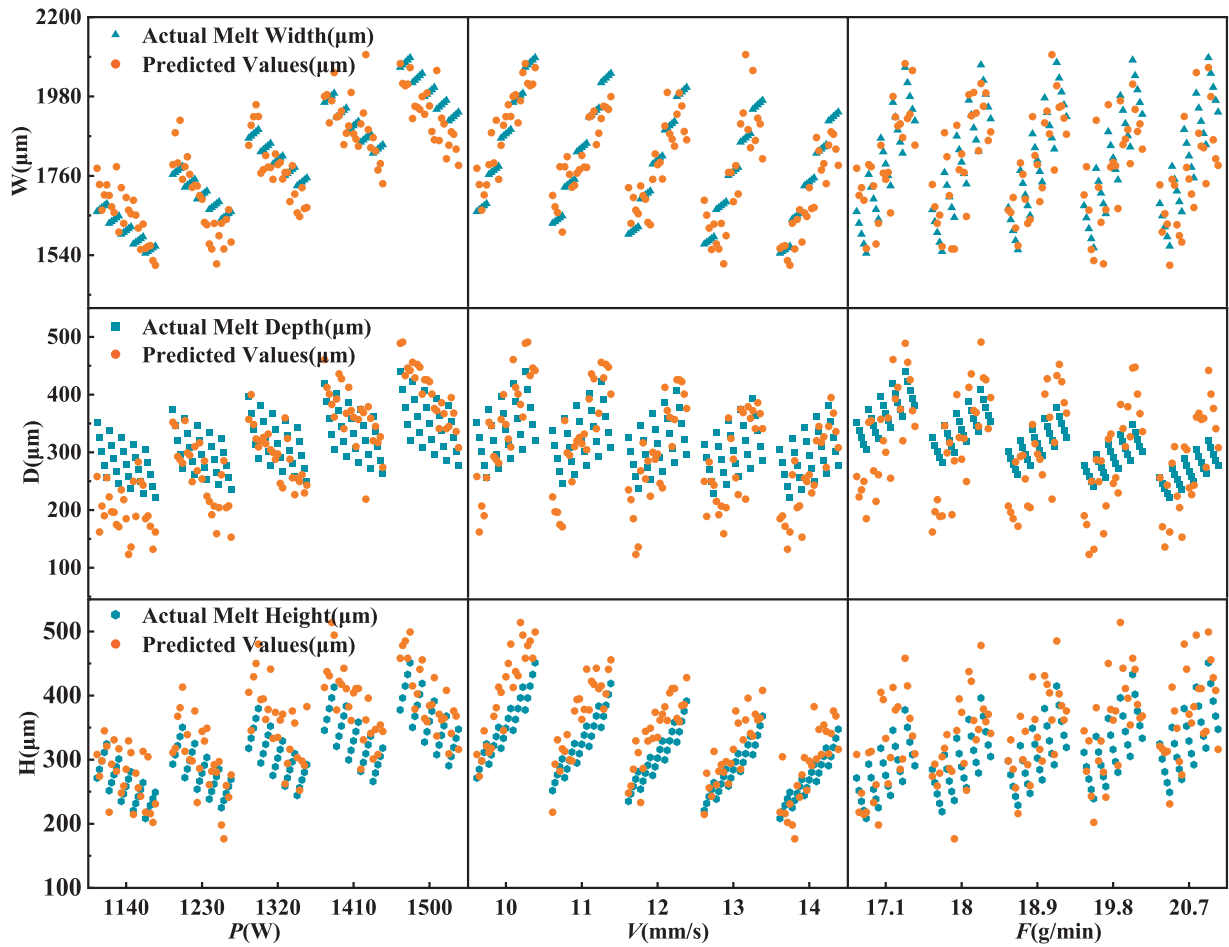


Fig. 8. Comparison between the predicted results of physical model and the actual values.

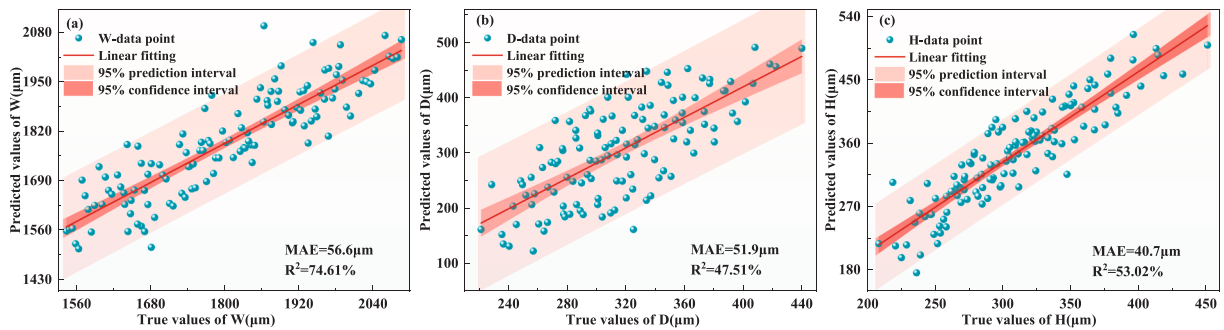


Fig. 9. Analysis of the predicted results of the physical model. (a) Width of melt pool; (b) depth of melt pool; (c) height of melt pool.

Table 5

Detailed parameters of the comparison methods.

Methods parameters	GSABO-BPNN	BPNN	SVM	RFR
Learning rate	0.001	0.001	/	/
Hidden layers	9	9	/	/
Dropout rate	0.1	0.1	/	/
Optimizer	Adam	Adam	/	/
Activation function	sigmoid	sigmoid	/	/
Kernel function	/	/	RBF	/
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network weights and thresholds. The SVM and RFR models adopt their standard regression implementations, with their model parameters and

structures determined through parameter searching on the training dataset.

For model performance evaluation, the mean absolute error (MAE) and the coefficient of determination (R^2) are selected as the primary evaluation metrics to quantitatively assess the prediction performance for W, D and H [47]. MAE is used to measure the average deviation between the predicted values and the experimental measurements, thereby providing an intuitive indication of prediction accuracy, whereas R^2 characterizes the model's ability to fit the overall variation trend of the experimental data. Lower MAE values and higher R^2 values indicate superior predictive performance. Since the calculations of MAE and R^2 are based solely on the error relationship between model outputs and ground truth values, independent of the model's internal structure or modeling form, these metrics provide a unified, objective, and

comparable evaluation criterion for models employing different multi-output modeling strategies. The mathematical expressions of MAE and R^2 are given as follows:

$$MAE = \frac{1}{n} \sum_{i=1}^N |y_i - \hat{y}_i| \quad (39)$$

$$R^2 = 1 - \frac{\sum_{i=1}^N (y_i - \hat{y}_i)^2}{\sum_{i=1}^N (y_i - \bar{y})^2} \quad (40)$$

To highlight the suitability of the models for multi-output prediction tasks, this study compares their performance when simultaneously predicting W, D and H. Within this framework, the role of the GSABO algorithm is to globally optimize the weights and bias parameters of the BPNN, thereby improving the overall convergence behavior, stability, and generalization capability of the network in multi-output prediction tasks. It should be noted that GSABO, as an optimization strategy, only operates on the parameter search process of the network and does not alter the mathematical nature or output formulation of the multi-output regression problem.

4.2.2. Performance validation of GSABO

To further validate the performance advantages of the proposed GSABO in global initialization optimization of BPNN parameters, as well as its stability and robustness in small-sample nonlinear modeling tasks, this study conducts a comparative analysis between GSABO and several representative intelligent optimization algorithms. Under the premise that the network architecture, data partitioning strategy, fitness function definition, and optimization objectives are kept completely identical, GSABO is compared with particle swarm optimization (PSO), genetic algorithm (GA), grey wolf optimizer (GWO), differential evolution (DE), and simulated annealing (SA). Considering the inherent stochasticity of intelligent optimization algorithms under random initialization, each algorithm is executed for 100 iterations under identical settings and independently repeated 30 times.

Table 6 summarizes the statistical results of the final fitness values obtained by the six optimization algorithms over 30 repeated experiments, including the minimum, maximum, mean, and standard deviation. It can be observed that GSABO achieves a minimum fitness value of $9.33\text{E-}4$, a maximum of $2.45\text{E-}3$, a mean value of $1.3\text{E-}3$, and a standard deviation of $3.97\text{E-}4$, all of which are lower than those of the other optimization algorithms, indicating its superior global optimization capability in obtaining high-quality solutions. By contrast, DE and SA exhibit relatively large performance fluctuations and are more sensitive to random initial conditions. PSO, GA, and GWO also fall short of GSABO in terms of solution quality and stability. Fig. 10 illustrates the box plots of the final fitness values and the fitness convergence curves. As shown in Fig. 10(a), GSABO exhibits a relatively narrow box range and a lower median value, indicating that most repeated runs achieve stable and high-quality solutions. Fig. 10(b) shows that all algorithms can rapidly reduce the fitness value in the initial stage; however, significant differences emerge during the later iterations. GSABO is able to continuously improve solution quality and converge to a lower fitness level, whereas DE and SA exhibit premature convergence stagnation, GWO shows relatively large fluctuations during convergence, and PSO attains a final accuracy slightly inferior to that of GSABO.

Table 6
Repeatability validation results of the compared optimization algorithms.

Optimizer	Min	Max	Mean	Std
GSABO	9.33E-4	2.25E-3	1.35E-3	3.97E-4
PSO	9.75E-4	2.77E-3	1.49E-3	4.79E-4
GA	1.07E-3	3.16E-3	1.56E-3	4.74E-4
GWO	9.68E-4	2.59E-3	1.49E-3	4.21E-4
DE	1.13E-3	3.35E-3	1.64E-3	4.91E-4
SA	1.07E-3	3.32E-3	1.72E-3	5.28E-4

In summary, for the small-sample and highly nonlinear laser cladding morphology prediction task investigated in this study, GSABO demonstrates clear advantages in terms of optimization accuracy, stability, and convergence performance. By globally optimizing the initial weights and thresholds of the BPNN, GSABO effectively reduces the model's dependence on random initialization and provides more favorable initial solutions for subsequent local training, thereby enhancing the overall predictive performance and generalization capability of the model.

4.3. Analysis of the comparison results

4.3.1. Data-driven machine learning models

To evaluate the performance of data-driven models in predicting the melt pool morphology of single-track layer, MAE and R^2 were used for analysis, with the results shown in Tables 7 and 8 and Fig. 11. In the training phase reported in Table 7, the GSABO-BPNN model achieved superior accuracy across all output dimensions (W, D, H) compared with BPNN, SVM, and RFR. For example, in predicting W, the MAE of GSABO-BPNN was $34.5\text{ }\mu\text{m}$, whereas those of BPNN, SVM, and RFR were higher at $40.5\text{ }\mu\text{m}$, $41.6\text{ }\mu\text{m}$, and $45.0\text{ }\mu\text{m}$, respectively. A similar trend was observed in the testing phase: the MAE of GSABO-BPNN for W was $38.2\text{ }\mu\text{m}$, clearly lower than $45.2\text{ }\mu\text{m}$ for BPNN, $43.5\text{ }\mu\text{m}$ for SVM, and $43.1\text{ }\mu\text{m}$ for RFR. These results indicate that GSABO-BPNN not only fits the training data well but also exhibits good generalization ability, thereby minimizing prediction errors.

The R^2 in Table 8 reflects the goodness of fit, and it can be seen that GSABO-BPNN outperforms the other models on both the training and test sets. In the training set, for predicting W, GSABO-BPNN achieved an R^2 of 90.61%, exceeding 84.80% for BPNN, 86.53% for SVM, and 83.75% for RFR. In the test set, the R^2 of GSABO-BPNN for predicting W was 76.42%. Although this is lower than 77.43% for BPNN, 80.59% for SVM, and 80.75% for RFR, GSABO-BPNN maintains a more stable and higher overall goodness of fit when considering the joint predictive performance across all outputs (W, D, H). In summary, optimizing the BPNN with the GSABO strategy delivers outstanding performance for predicting the melt pool morphology of single-track cladding layer. It surpasses conventional models of BPNN, SVM, and RFR in both error minimization and goodness of fit. However, the data-driven GSABO-BPNN attains an average R^2 of only 81.04% on the test set, indicating room for improvement in predictive accuracy. Therefore, it is necessary to integrate the physical model with GSABO-BPNN to construct the process parameters to melt pool morphology mapping model with higher predictive accuracy.

4.3.2. Physics-machine learning hybrid model

The integration of physical models with machine learning models aims to further correct the prediction deviations introduced by simplified assumptions or unmodeled factors, building upon the mapping relationship between process parameters and clad layer morphology already captured by the physical model. The comparative results of different physics-machine learning hybrid models in terms of MAE and R^2 are presented in Tables 9 and 10 and Fig. 12.

In terms of the MAE metric, the physics-GSABO-BPNN model exhibits an overall advantage in prediction accuracy, which is particularly evident in the testing phase. During the training phase, for W, the MAE of the physics-GSABO-BPNN model is comparable to those of other hybrid models; however, for the prediction of D and H, its MAE is significantly lower than that of the physics-BPNN, physics-SVM, and physics-RFR models. In the testing phase, the physics-GSABO-BPNN model maintains the lowest MAE across all output parameters (W, D, and H). Specifically, the MAE for W is $37.1\text{ }\mu\text{m}$, which is markedly lower than those of the physics-BPNN ($53.6\text{ }\mu\text{m}$), physics-SVM ($40.1\text{ }\mu\text{m}$), and physics-RFR ($44.3\text{ }\mu\text{m}$) models, with consistent superiority also observed for D and H. The consistently low MAE observed on the test set indicates that, under the incorporation of physical constraints, the GSABO-BPNN

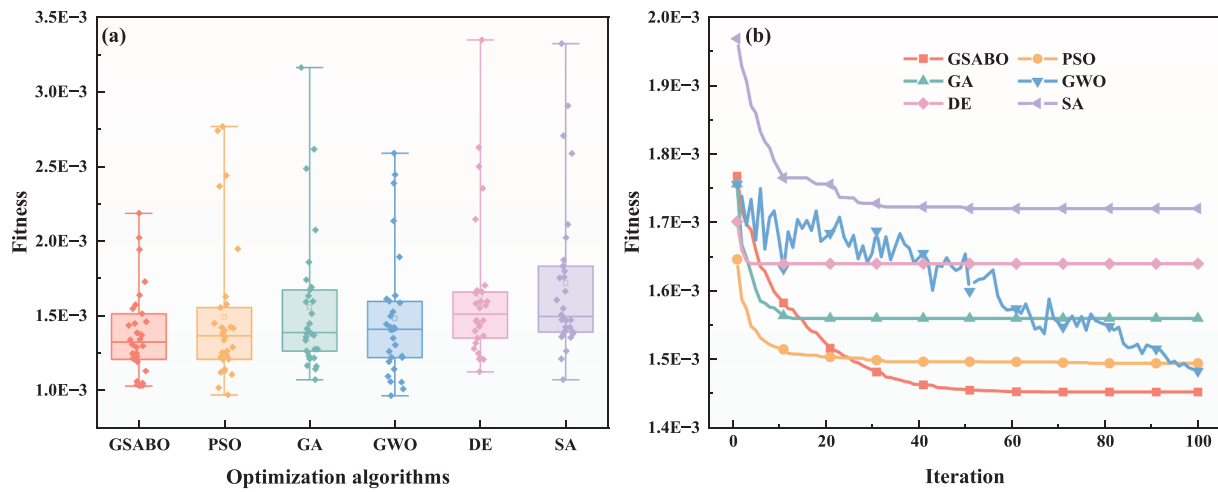


Fig. 10. Comparative validation results of optimization algorithms. (a) Box plots of repeatability validation results; (b) fitness convergence curves of the algorithms.

Table 7

The MAE of data-driven machine learning models.

Dataset	Output	GSABO-BPNN	BPNN	SVM	RFR
Training	W (μm)	34.5	40.5	41.6	45.0
	D (μm)	24.0	28.2	24.8	29.4
	H (μm)	20.3	22.1	21.4	26.1
Testing	W (μm)	38.2	45.2	43.5	43.1
	D (μm)	24.2	39.5	32.8	27.8
	H (μm)	21.5	27.5	25.4	28.1

Table 8

The R^2 of data-driven machine learning models.

Dataset	Output	GSABO-BPNN	BPNN	SVM	RFR
Training	W	90.61%	84.80%	86.53%	83.75%
	D	87.78%	84.01%	87.32%	82.86%
	H	87.39%	86.34%	87.76%	81.52%
Testing	W	76.42%	77.43%	80.59%	80.75%
	D	80.93%	65.14%	74.53%	78.81%
	H	85.76%	61.54%	67.53%	52.17%

model can effectively correct the systematic biases of the physical model and maintain stable prediction accuracy under unseen process conditions, thereby demonstrating strong generalization capability.

In terms of the R^2 , the physics-GSABO-BPNN model likewise exhibits a significant advantage during the training phase. Specifically, the R^2 values for predicting W, D, and H reach 94.51%, 90.69%, and 93.11%, respectively, all of which are higher than those achieved by the other physics-machine learning hybrid models. These results indicate that, during training, the incorporation of physical prior information enables GSABO-BPNN to more effectively exploit the existing empirical constraints between process parameters and melt pool morphology, thereby achieving a more stable fitting performance in a statistical sense.

The R^2 value in the testing phase provides a more representative measure of the model's generalization capability. On the testing set, the physics-GSABO-BPNN model achieves R^2 values of 92.09%, 86.58%, and 92.46% for W, D, and H, respectively. These values are not only significantly higher than those of the physics-BPNN, physics-SVM, and physics-RFR models, but also exhibit a smaller gap between the training and testing R^2 . This indicates that, under physical constraints, the model possesses more robust transferability to previously unseen combinations of process parameters. By contrast, the other physics-machine learning hybrid models exhibit varying degrees of R^2 degradation in the testing phase, suggesting that insufficient parameter optimization or suboptimal coupling strategies between physical models and conventional

machine learning methods can still limit generalization performance.

A further comparison between the physics-machine learning hybrid models and the purely data-driven models reveals that the incorporation of physical models leads to overall improvements in prediction performance across all hybrid approaches. Taking GSABO-BPNN as an example, its MAE in the testing phase is reduced by an average of 6.85%, while its R^2 is increased by an average of 11.79%. For traditional models such as BPNN, SVM, and RFR, although the corresponding hybrid models exhibit relatively limited improvements in MAE, the introduction of physical constraints results in more stable response behaviors of the predictions across the process parameter space. This reduces excessive dependence on the training data distribution and effectively suppresses non-physical anomalous predictions, thereby yielding superior overall goodness of fit compared with purely data-driven models.

In summary, by integrating physical models with machine learning models, the hybrid approaches achieve significant improvements over purely data-driven models in terms of prediction accuracy, fitting stability, and generalization capability to unseen process conditions. Among the various physics-machine learning hybrid models, the physics-GSABO-BPNN consistently demonstrates more stable and superior performance across all evaluation metrics, indicating its higher engineering applicability and reliability for single-track laser cladding morphology prediction.

4.4. The interpretability analysis of SHAP

4.4.1. Data-driven machine learning model

The importance contributions of different process parameters (P , V , F) to predicting melt pool morphology characteristics (W, D, H) in the data-driven model are shown in Fig. 13. The SHAP analysis indicates that parameter sensitivities differ across models, and the contribution of the same process parameter varies from one model to another. In the GSABO-BPNN model, the parameter importance ranking is $P > V > F$. Among them, P most strongly affects W and D, whereas F primarily manifests in the prediction of H. The BPNN model exhibits a similar parameter importance ranking pattern, though the distinction among parameter contributions is somewhat weaker. By contrast, the importance ranking for both SVM and RFR becomes $V > F > P$. These differences stem from disparities in model architectures and feature sensitivities. Neural-network models place greater emphasis on energy input, thereby highlighting the dominant role of P . Models based on kernel functions or tree structures are more sensitive to variations in V . Notably, the feature contribution pattern of GSABO-BPNN aligns more closely with the fact physical process, where P directly determines energy input and thus governs melt pool morphology, conferring stronger

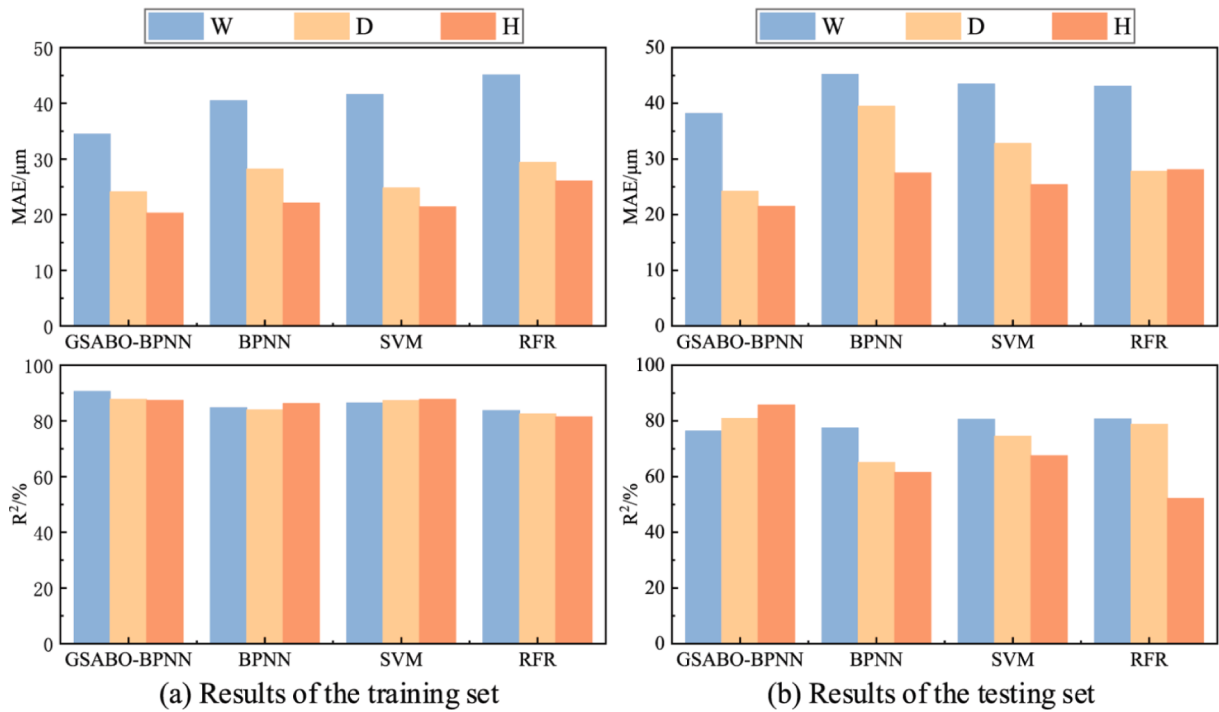


Fig. 11. Comparison of predictive performance based on data-driven machine learning models.

Table 9

MAE of the physics-machine learning hybrid model.

Dataset	Output	GSABO-BPNN	BPNN	SVM	RFR
Training	W (μm)	36.7	36.4	32.5	36.5
	D (μm)	21.4	26.4	20.2	27.8
	H (μm)	16.2	21.4	17.5	20.1
Testing	W (μm)	37.1	53.6	40.1	44.3
	D (μm)	24.4	27.3	30.2	32.4
	H (μm)	17.8	33.9	24	23.5

Table 10

R² of the physics-machine learning hybrid models.

Dataset	Output	GSABO-BPNN	BPNN	SVM	RFR
Training	W	94.51%	89.19%	88.06%	87.76%
	D	90.69%	85.01%	88.67%	84.31%
	H	93.11%	85.88%	89.02%	87.78%
Testing	W	92.09%	73.11%	82.85%	81.84%
	D	86.58%	76.38%	76.48%	78.43%
	H	92.46%	83.16%	70.87%	81.15%

interpretability[48].

In summary, P and V are the primary governing factors for melt pool morphology (W , D , H), although their relative importance depends on the specific model. The BPNN model optimized by the GSABO strategy aligns more closely with process patterns in feature ranking, providing a more reliable basis for understanding the mechanism of the laser cladding process and achieving process parameter control.

4.4.2. Physics-machine learning hybrid model

In the physics-machine learning hybrid model, the importance contributions of different process parameters (P , V , F) to the prediction residuals of melt pool morphology (W , D , H) are shown in Fig. 14. Laser cladding is a complex process with multi-physical fields. The distribution of prediction residuals of the melt pool morphology under different process parameters actually reflects the leading sources of factors not fully considered in the physical model. This figure reveals significant

differences in feature importance across different models, indicating variations in their adaptability regarding the learning of prediction residuals and the capture of physical mechanisms.

In the physics-GSABO-BPNN model, the feature importance ranking is $F > V > P$. The mean absolute SHAP value for F is the highest, indicating that the physical model inadequately accounts for the influence of F on W and D . Consequently, prediction residuals primarily rely on compensation through F to correct the predicted values of W and D . This is primarily because attenuation of the laser through the powder stream and the partition of laser energy between powder and substrate are complex, which a simple physical model cannot fully capture. The mean absolute SHAP value of V ranks second, reflecting insufficient simplifications in the physical model regarding energy interaction time and melt pool dynamics, which induces systematic bias in morphology prediction. The contribution of P is least significant, indicating that the physical model has already reasonably captured the dominant effect of energy input. Through global optimization and nonlinear fitting capability, the physics-GSABO-BPNN model effectively extracts discrepancies arising from unconsidered factors. Its feature importance ranking closely matches the neglected mechanisms, underscoring sound interpretability.

By contrast, the physics-BPNN model also exhibits the trend $F > V > P$, but its ability to differentiate the parameters within the prediction residuals is weak, indicating limited capture of unconsidered factors. The physics-SVM model shows P as the most important, indicating that its kernel mapping treats energy input as the source of prediction residuals, which deviates from the actual physical interpretation. In the physics-RFR model, the contributions of P , V , and F are all low, and the effect of V is markedly weakened, indicating under fitting of the tree model when learning the complex nonlinear structure of the residuals. By comparison, the GSABO-BPNN model can reasonably identify the influences of factors neglected by the physical model and correct them to obtain more accurate predictions.

In summary, the physics-machine learning hybrid model not only achieves complementary strengths between physical mechanisms and data-driven approaches but also explains the shortcomings of the physical model by feature contribution analysis of prediction residuals.

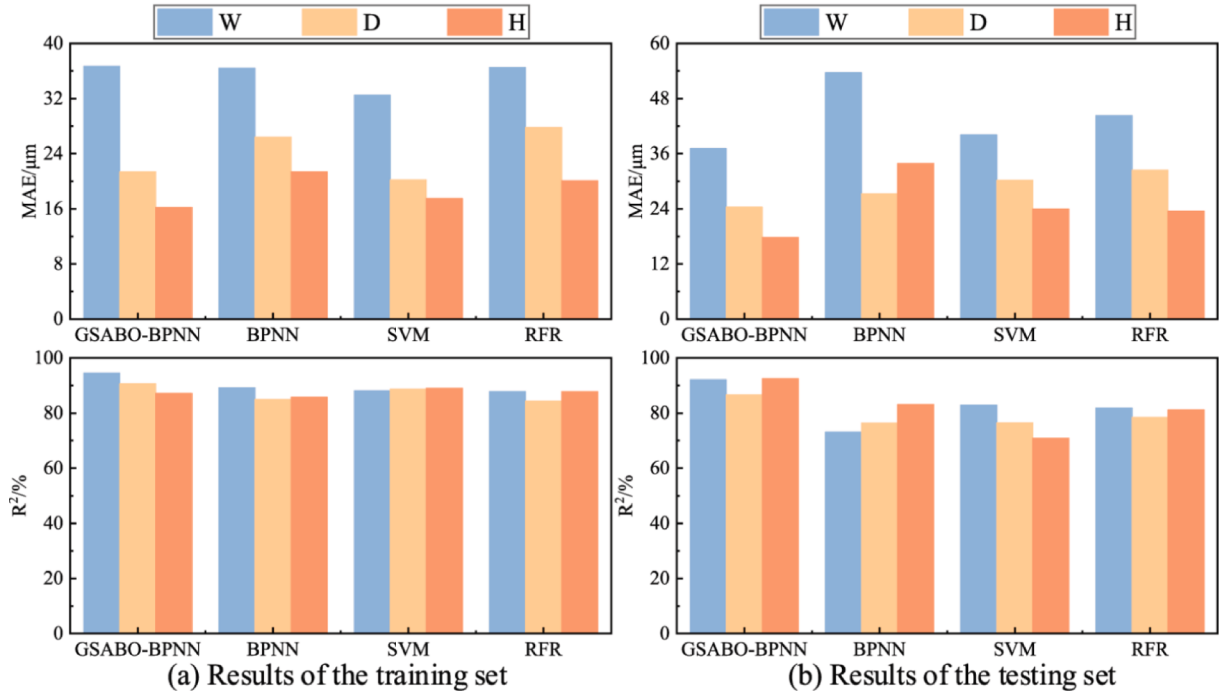


Fig. 12. Comparison of prediction performance based on hybrid physics-machine learning models.

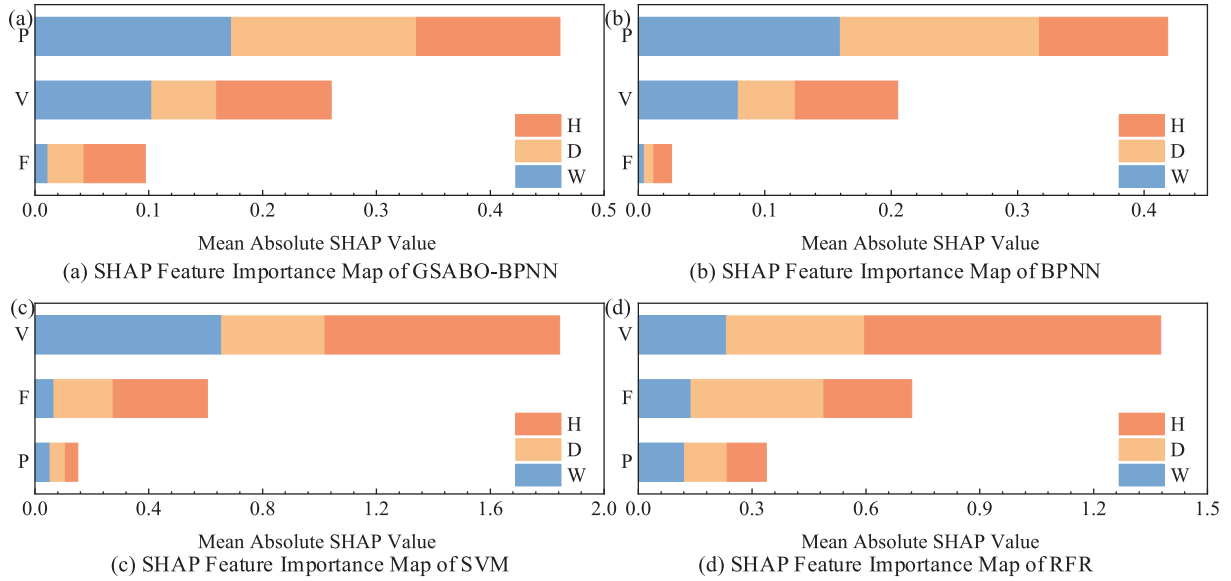


Fig. 13. SHAP feature importance map based on a data-driven machine learning models.

Through optimized search and nonlinear fitting capabilities, the GSA-BO-BPNN model more accurately captures and corrects these unconsidered factors, thereby markedly improving prediction accuracy while preserving physical plausibility.

4.5. Physics-machine learning hybrid model of transfer learning

To further validate the predictive performance of the constructed physics-machine learning hybrid model under small-sample dataset conditions, this study introduces a transfer learning strategy, with the data-driven GSABO-BPNN model serving as a comparison baseline. Given the limited sample size of the target-domain dataset, the performance evaluation of the transfer learning models follows the same repeated random data partitioning strategy described earlier, in order to

reduce the influence of random data splitting on the evaluation results. The predictive results and performance of the different models are shown in Tables 11 and 12. A visual illustration of the predictions obtained by the two methods is shown in Fig. 15. Overall, the transfer learning (TL) based physics-machine learning hybrid model delivers significantly better predictions across the W, D, and H dimensions than the data-driven machine learning (ML) model under small-sample conditions.

The ML model exhibits generally high prediction errors with substantial variability. For some samples, the error exceeds 20% and even surpasses 30%, which severely undermines the practicality of the model. Under the same testing conditions, the predictions of the TL model are closer to the ground truth, with errors kept within 10%, markedly reducing predictive uncertainty. This advantage indicates that transfer

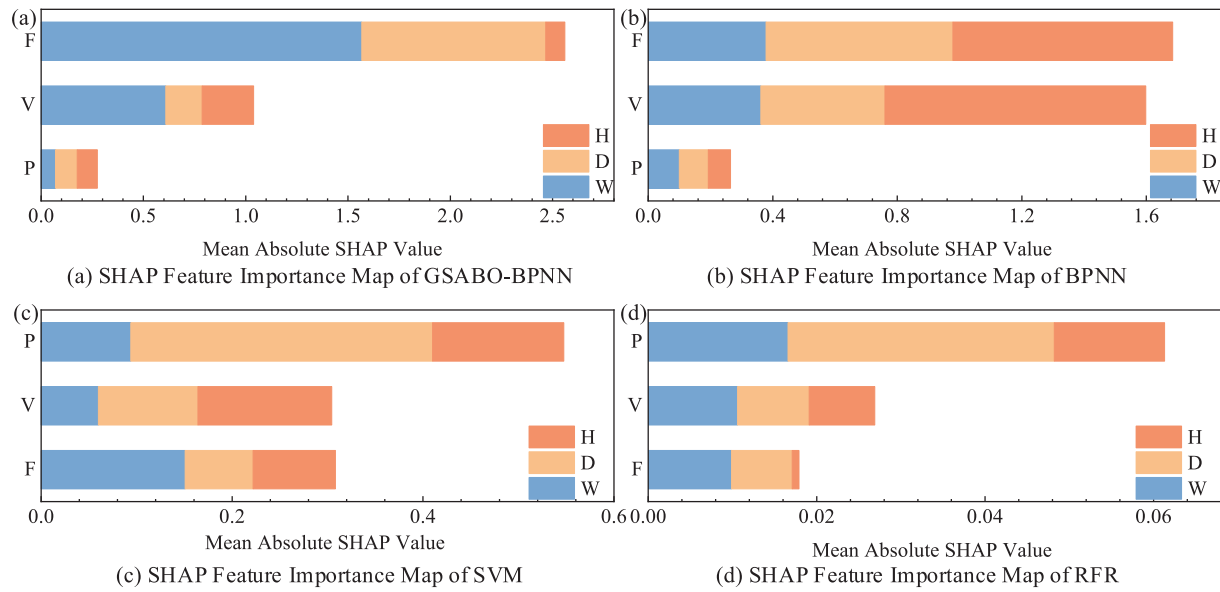


Fig. 14. SHAP feature importance map based on a physics-machine learning hybrid models.

Table 11

Comparison of the test set data based on ML and TL with the actual values.

Output	NO.	Actual values/ μm	ML/ μm	Error	TL/ μm	Error
W	1	2432.6	2643.2	8.65%	2301.5	5.38%
	2	2691.8	2953.8	9.73%	2805.2	4.23%
	3	3087.5	3108.9	0.69%	3120.7	1.07%
	4	2681.7	2825.0	5.34%	2671.0	0.40%
	5	2831.2	3039.6	7.36%	2749.9	2.87%
D	1	937.22	864.38	7.77%	898.42	4.14%
	2	551.02	491.70	10.76%	566.63	2.83%
	3	732.92	663.69	9.44%	693.52	5.37%
	4	485.16	383.57	20.93%	525.46	8.30%
	5	567.8	728.86	28.36%	615.04	8.31%
H	1	416.77	453.17	8.73%	450.7	7.97%
	2	1035.00	1186.20	14.60%	1035.2	0.02%
	3	844.83	755.11	10.62%	902.39	6.81%
	4	813.20	1084.00	33.33%	868.59	6.83%
	5	937.02	978.08	4.38%	979.51	4.53%

Table 12

The MAE and R^2 based on ML and TL.

Methods	MAE/ μm			-	R^2		
	W	D	H		W	D	H
ML	169.14	92.81	98.28		22.5%	62.3%	51.5%
TL	73.94	56.27	36.91		83.2%	90.8%	96.0%

learning, by incorporating the laws and constraints of the physical model and leveraging cross-domain knowledge transfer, can effectively mitigate distribution shift induced by small-sample datasets, enabling the model to retain strong generalization under limited data conditions.

The MAE and R^2 in Table 11 further corroborate this conclusion. For the ML model, the MAEs for W, D, and H are 169.14 μm , 92.81 μm , and 98.28 μm , respectively. By contrast, the TL model reduces them markedly to 73.94 μm , 56.27 μm , and 36.91 μm , with an average MAE reduction exceeding 40%. The R^2 of the ML model is low, scarcely capturing the actual trends. Compared with the ML model, the TL model attains R^2 values of 0.832, 0.908, and 0.960 for the three outputs, indicating that the method fits the data effectively and maintains strong generalization under small-sample and cross-domain conditions.

In summary, the transfer learning based physics-machine learning hybrid model demonstrates significant advantages in predicting melt

pool morphology, particularly on small-sample datasets. Compared with the machine learning model, this approach maintains lower MAE and higher R^2 under small-sample conditions, while offering adaptability across materials and equipment. These results provide compelling evidence for the effectiveness of transfer learning in addressing small-sample challenges and improving cross-domain modeling accuracy.

4.6. Optimize performance analysis

The optimization results and validations for the data-driven machine learning model and the transfer learning based physics-machine learning hybrid model are shown in Table 13 and Fig. 16. All experimental verification results were repeated three times and averaged. One set of verification results is shown in Fig. 17. As shown in Table 13, the ML model combined with NSGA II (ML-NSGA II) attains reasonably good predictions during optimization, but performs poorly in the experimental validation stage. Although the error for W remains within a reasonable range, the experimental errors for D and H reach 121.27% and 20.64%, respectively, far exceeding acceptable levels for engineering applications. This indicates that under small-sample conditions, the data-driven ML model may exhibit the phenomenon of “good optimization yet large experimental error”. Such a model overfits the limited data distribution and is incapable of accurately capturing the complex physical phenomena in laser cladding, thus producing unreliable predictions. By contrast, the optimization and validation outcomes of the TL model combined with NSGA II (TL-NSGA II) are more stable. The optimized results align closely with the target values, and in experimental validation the errors for W, D, and H are constrained to 1.79%, 7.14%, and 6.50%, respectively, markedly outperforming ML-NSGA II. In summary, data-driven machine learning suffers from insufficient accuracy and poor experimental generalization on small-sample datasets. By contrast, an optimization framework that combines the physics-machine learning hybrid model with transfer learning achieves dual consistency between prediction and experiment, exhibiting greater robustness and engineering value.

5. Conclusions

Focusing on the prediction and optimization of the melt pool morphology of single-track layers in laser cladding, this study proposes a modeling framework that integrates physical model with machine learning. By leveraging transfer learning technology, the framework

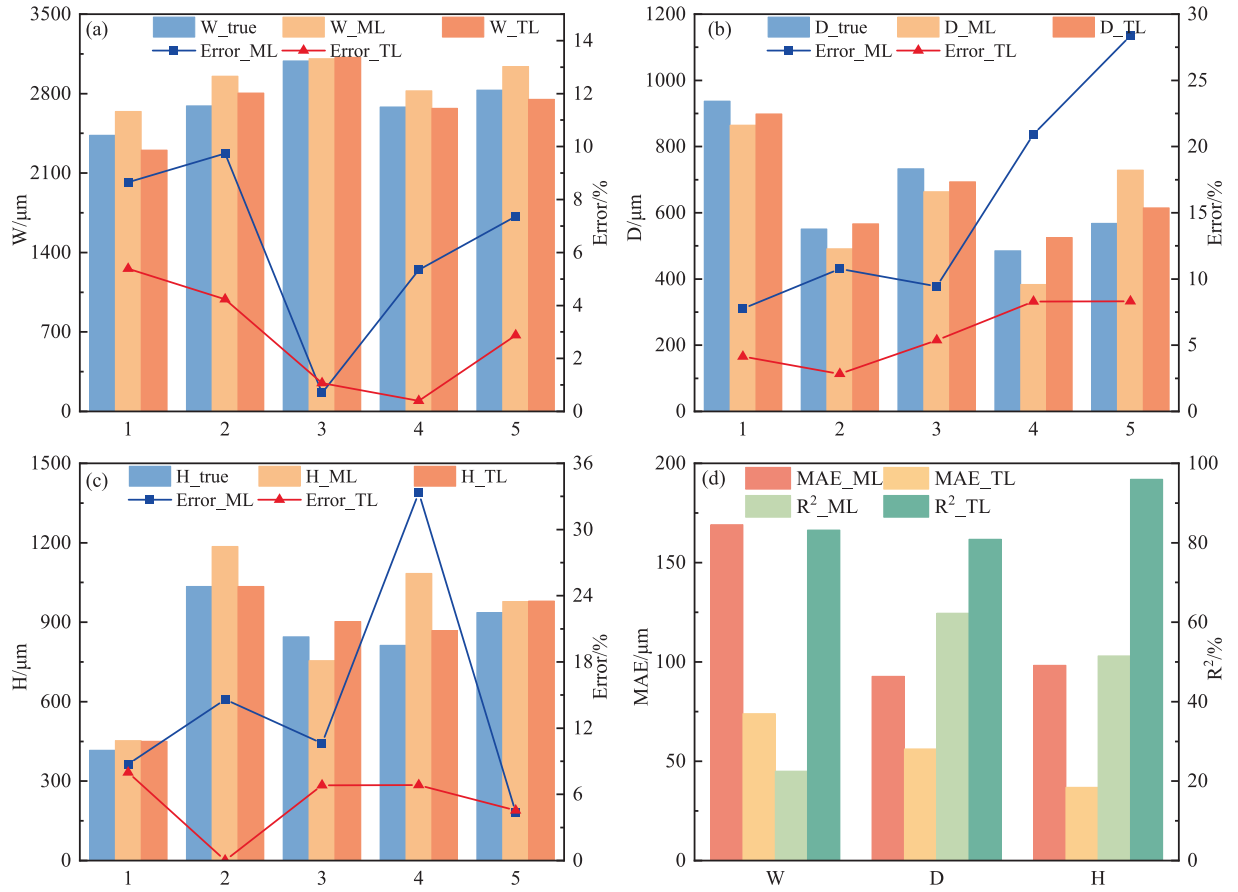


Fig. 15. Results of the prediction model based on ML and TL. (a) The results of W; (b) the results of D; (c) the results of H; (d) MAE and R^2 of prediction result.

Table 13

Comparison of optimization results and verification based on ML-NSGA II and TL-NSGA II.

Methods	Process parameters (P, V, F)	—	W/ μm	D/ μm	H/ μm
—	—	Objective	3000	430	1000
ML-NSGA II	(1947.5 W, 9.04 mm/s, 13.04 g/min)	Optimized	2908.89	427.45	1049.03
		Error1	3.04%	0.59%	4.90%
		Experimental	2946.61 (± 27.86)	951.44 (± 64.29)	798.36 (± 38.55)
		Error1	1.78%	121.27%	20.64%
TL-NSGA II	(1862.8 W, 9.18 mm/s, 16.34 g/min)	Optimized	3005.21	436.52	1055.21
		Error2	0.17%	1.52%	5.52%
		Experimental	3053.62 (± 58.95)	460.72 (± 41.80)	1065.01 (± 61.84)
		Error2	1.79%	7.14%	6.50%

enables melt pool morphology prediction and process parameter optimization under the conditions of cross-material, cross-equipment, and non-overlapping process parameter ranges. The specific conclusions are as follows.

- (1) The physical model proposed in this study, which combines the Rosenthal heat source with a Gaussian heat source, can adequately describe the relationship between process parameters and the melt pool morphology of the clad layer in laser cladding. The prediction trends of the model for W, D, and H are basically consistent with the experimental results, confirming its effectiveness as a physical prior model. However, due to its insufficient consideration of the thermal properties of materials and the complex coupling mechanisms within the melt pool, there are

certain deviations in the prediction results, which limits its applicability under more complex working conditions.

- (2) A hybrid model integrating the physical prior with a GSABO-BPNN was developed. This hybrid model exhibits higher predictive accuracy, greater robustness, and better generalization than data-driven models. The “physical prior + data-driven” strategy effectively reduces overfitting risks while enhancing predictive precision.
- (3) An interpretability analysis of the prediction models was performed using the SHAP method. The results show that, among data-driven models, the GSABO-BPNN model exhibits a ranking of feature importance contributions of process parameters to melt pool morphology as $P > V > F$, which is more consistent with physical reality. In the hybrid model, the contribution ranking of process parameters is $F > V > P$. This indicates that the physical model inadequately describes the relationships between F , V , and melt pool morphology, but can effectively capture the main effect of P . The results not only clarified the physical sources of residuals in the physical model but also provided a basis for subsequent improvements to the physical model.
- (4) Through the physical prior model and transfer learning strategy, the prediction model can still maintain high predictive accuracy under cross-material, cross-equipment, and conditions with differences in process parameter distributions, significantly reducing its reliance on experimental data. Furthermore, a multi-objective optimization model was established by integrating NSGA II, verifying the effectiveness of the proposed method in terms of prediction and optimization performance.

In summary, this study proposes an interpretable physics-based hybrid modeling method combined with the improved BPNN. Relying

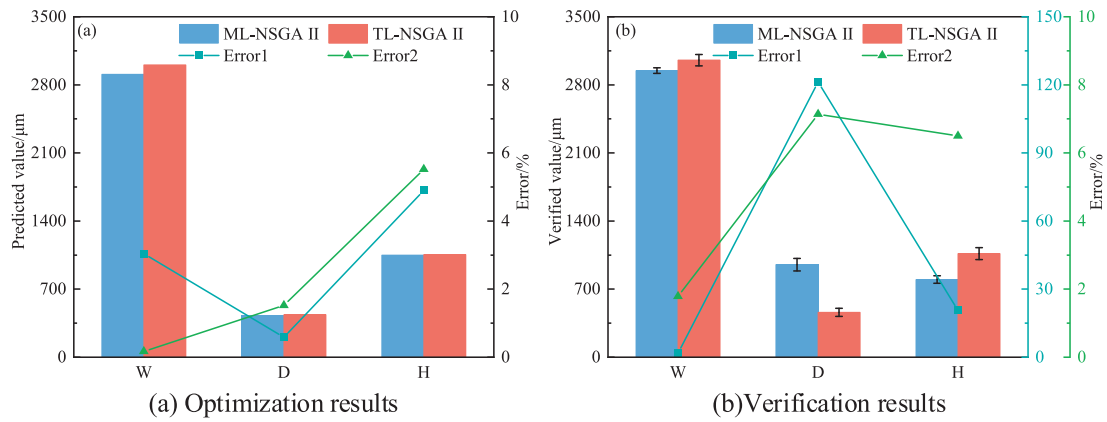


Fig. 16. Comparison of optimization results and validation based on ML-NSGA II and TL-NSGA II. (a) Optimization results; (b) verification results.

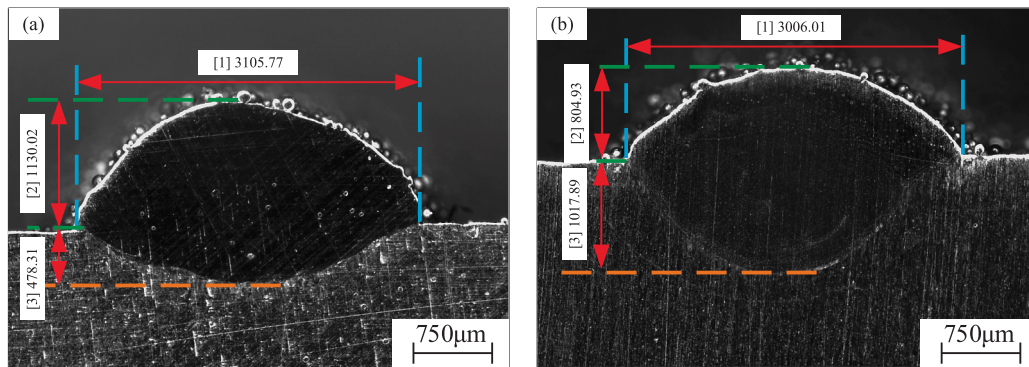


Fig. 17. Verification results based on ML-NSGA II and TL-NSGA II. (a) Verification results of TL-NSGA II; (b) verification results of ML-NSGA II.

on transfer learning strategies and intelligent optimization algorithms, this method demonstrates application potential in cross-domain process prediction and process parameters regulation. In future research, on the one hand, the research object can be extended to multi-track lap or multi-layer laser cladding processes, while integrating multi-physics field coupling mechanisms to construct a more comprehensive physical model for multi-track/multi-layer laser cladding layers, thereby further improving the accuracy and adaptability of the physical model. On the other hand, the modeling method can be combined with real-time monitoring and feedback control technologies to achieve intelligent and adaptive regulation of the laser cladding process, providing more reliable and efficient technical support for engineering applications.

CRediT authorship contribution statement

Yunchuan Peng: Writing – original draft, Validation, Methodology,

Investigation, Data curation, Conceptualization. **Tao Li:** Writing – review & editing, Resources, Project administration, Funding acquisition. **Yanbin Du:** Supervision, Resources, Investigation. **Weiwei Liu:** Supervision, Funding acquisition. **Huanqiang Liu:** Supervision, Resources. **Zhijie Zhou:** Validation, Resources, Investigation.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Derivation of correction factors and selected parameters

The derivation processes of the correction factors and selected parameters involved in the physical model development presented in Section 3 are provided in Appendices A.1–A.3. The schematic illustration of the corrections is shown in Fig. A.1, and the specific parameters of the physical model are listed in Table A.1.

A.1 Powder stream attenuation model and temperature calibration coefficient

In powder-fed laser cladding, the powder stream is regarded as a medium that absorbs and scatters the incident laser radiation. According to the Beer-Lambert law, the transmitted intensity of laser radiation in an absorbing medium decays exponentially with the optical thickness. For a given nozzle configuration and optical path length, the effective optical thickness of the powder stream increases with increasing powder feed rate F . Considering the nonlinear variations in powder stream morphology and concentration within the experimental parameter range, the effective optical

thickness is approximated as a second-order function of F , thereby yielding a semi-empirical attenuation model:

$$P_{at} = P[\exp(-(k_1 F + k_2 F^2))] \quad (A1)$$

where P is the laser output power, P_{at} denotes the laser power reaching the melt pool region after passing through the powder stream, and k_1 and k_2 characterize the first-order and second-order attenuation effects associated with the powder feed rate F , respectively.

Considering the different absorption mechanisms of laser energy by the substrate and the powder, the substrate absorptivity and the powder utilization rate are introduced, and the effective heat input power is expressed as:

$$\begin{cases} P_w = \beta_a P_{at} + \eta_p (P - P_{at}) \\ \beta_a = \beta_w [1 + a_w \tan^{-1}(2h_0/r_b)] \end{cases} \quad (A2)$$

where η is the powder utilization rate, β_a is the substrate absorptivity considering the Brewster effect, β_w is the intrinsic absorptivity of the substrate, a_w is the Brewster coefficient; h_0 denotes the laser incident height, and r_b is the laser spot radius.

Based on the moving heat source temperature field, under the lateral isothermal contour condition of the melt pool, the physical model for the W can be expressed in the following implicit equation form (Eq. (7) in the main text):

$$W = \frac{C_w}{\sqrt{\frac{vW}{4\alpha} + 1}}, C_w = \frac{nP_w}{\pi e k (T_m - T_0)} \quad (A3)$$

where v is the scanning speed, α is the thermal diffusivity, and k , ρ , C_p , T_m , and T_0 denote the physical properties and temperature parameters of the material, respectively. The specific parameters are listed in Table A.1. The parameter n is a temperature calibration coefficient introduced to compensate for the deviation between the ideal heat source model assumptions and the actual heat transfer process.

Since the governing equation for the melt pool width W is implicit, a fixed-point iteration method is adopted for numerical solution. For the i -th set of experimental parameters, the iterative scheme at the t -th iteration step can be expressed as:

$$W_i^{(t)} = \frac{C_{w,i}}{\sqrt{\frac{v_i W_i^{(t-1)}}{4\alpha} + 1}} \quad (A4)$$

When the difference between two successive iteration results satisfies the convergence criterion, the iteration is considered converged, yielding the predicted melt pool width under the given parameter set. Given an initial value of the temperature calibration coefficient n , the parameters k_1 and k_2 enter the melt pool width model only through $P_{at,j}(k_1, k_2)$ and $P_{w,j}(k_1, k_2)$, resulting in the predicted melt pool width. In this study, the sum of squared residuals of the melt pool width is taken as the objective function, and nonlinear least-squares inversion is performed to identify k_1 and k_2 :

$$\min_{k_1, k_2} \sum_{i=1}^N (W_i^{phy}(k_1, k_2, n) - W_i^{exp})^2 \quad (A5)$$

where W_{exp} denotes the experimentally measured melt pool width.

When constructing the physical model based on the analytical temperature field of a moving heat source, the required physical properties, such as thermal conductivity, specific heat capacity, and density, are taken as typical values of the material at room temperature. It should be noted that during the laser cladding process, the material undergoes rapid heating, melting, and cooling, during which its physical properties vary with temperature [49]. If constant room temperature properties are directly employed, certain deviations may be introduced in melt pool scale predictions. To compensate for the temperature-dependent property variations and complex heat transfer phenomena within the melt pool, such as convection and phase transformation, that are not explicitly considered in the analytical model, a temperature calibration coefficient n is introduced. This coefficient serves to apply a global scaling correction to the energy-temperature relationship, thereby improving the prediction accuracy and engineering applicability of the analytical model under practical laser cladding conditions.

Considering that the temperature calibration coefficient n plays a scaling role in melt pool width prediction, n is further treated as an outer-loop optimization parameter in this study. Within a reasonable range, n is searched, and for each candidate value of n , the aforementioned inversion procedures for k_1 and k_2 are repeated. The coefficient of determination of the predicted melt pool width is then computed over the dataset as:

$$R^2 = 1 - \frac{\sum_i (W_i^{phy} - W_i^{exp})^2}{\sum_i (W_i^{exp} - \bar{W}_i^{exp})^2} \quad (A6)$$

The parameter set (k_1, k_2, n) that yields the maximum R^2 value (equivalently, the minimum sum of squared residuals of the melt pool width) is finally selected as the optimal model parameter set, with the detailed results summarized in Table A.1. As shown in Fig. A.1(a and b), when $n = 0.8$, the residuals of the melt pool width prediction exhibit an overall negative shift, with a median residual of approximately -0.16 , indicating a systematic underestimation of the melt pool width under this condition. When the temperature calibration coefficient is adjusted to $n = 0.9$, the goodness of fit of the predicted melt pool width is significantly improved, with an R^2 value of 0.746 (as shown in Fig. 9). Meanwhile, the median of the residual distribution approaches zero, indicating that the predictions are statistically consistent with the experimental measurements.

The above results demonstrate that by introducing and appropriately calibrating the temperature calibration coefficient n , the systematic bias arising from the use of room temperature physical properties can be effectively corrected. This enables the analytical physical model to more accurately represent the equivalent heat transfer behavior in the actual cladding process while retaining its interpretability. Therefore, the room temperature material properties adopted in this study are not directly used for uncorrected temperature prediction. Instead, they form an equivalent physical description under the constraint of the temperature calibration coefficient, thereby ensuring the credibility of the model predictions at the engineering application level.

A.2 Residual-ratio regression analysis of the depth correction factor.

The main text provides the predicted melt pool depth without incorporating any correction factor is given by Eq. (9). Experimental statistical analysis indicates that the powder feed rate F exerts a secondary yet non-negligible modulating effect on melt pool depth. Since the heat source model is explicitly dependent only on P and V , correlation analysis nevertheless reveals a weak negative correlation (-0.21) between the F and melt pool depth D . If the influence of F is neglected, the discrepancy between the predicted and actual values of D will progressively increase with increasing F . To account for this effect, a semi-empirical correction factor is introduced in Eq. (10). The residual ratio of the melt pool depth is then defined as:

$$r_D(F) = \frac{D^{exp}}{D^{phy}(P, V)} = f(F) \quad (A7)$$

Taking the natural logarithm yields a linear regression form:

$$-\ln r_D = k_3 F + k_4 F^2 \quad (A8)$$

Let $y_i = -\ln(D^{exp}_i/D^{phy}_i(P, V))$ and $x_i = [F_i, F_i^2]^T$ be defined accordingly; then the parameter vector $[k_3, k_4]^T$ can be analytically determined using the ordinary least squares (OLS) method as:

$$\begin{bmatrix} k_3 \\ k_4 \end{bmatrix} = (\mathbf{X}^T \mathbf{X})^{-1} \mathbf{X}^T \mathbf{y} \quad (A9)$$

where $\mathbf{X} = [x_1^T, \dots, x_N^T]^T$ and $\mathbf{y} = [y_1, \dots, y_N]^T$. Fig. A.1(c) illustrates the distribution of the residual-ratio regression and the corresponding fitted curve. The resulting parameter set (k_3, k_4) obtained from the regression is summarized in Table A.1.

A.3 Regression analysis of the height correction factor

Based on the mass conservation principle and the parabolic cross-section assumption, the baseline model for the melt pool height is given by Eq. (14). As analyzed in Section 2.2.2 (as shown in Fig. 2), when the F is kept constant, an increase in the input energy density E_d leads to an increase in the H , indicating that more powder participates in the melting and deposition process and that the effective powder utilization is enhanced. Therefore, in practical processes, the effective powder utilization may vary with laser power due to mechanisms such as melting efficiency, spattering, and powder capture, leading to a systematic dependence of the melt pool height on the P . To characterize this effect, a power correction factor is given by Eq. (15).

To eliminate the influence of scanning speed, powder feed rate, and other factors on the melt pool height, a normalized form of H is defined as:

$$Z = \frac{H^{phy} v W}{F} = \frac{\eta_p}{40\rho} f(P) \quad (A10)$$

Thus, Z is proportional to $f(P)$, and taking the natural logarithm yields:

$$\begin{cases} \ln Z = a_0 + \beta_H \left(\frac{P}{P_0} - 1 \right) \\ a_0 = \ln \left(\frac{\eta_p}{40\rho} \right) \end{cases} \quad (A11)$$

where a_0 is determined by all constant terms independent of laser power, including geometric factors, unit conversion constants, powder utilization rate, and material density. Let $x_i = P_i/P_0 - 1$ and $y_i = \ln Z_i$. Fig. A.1(d) shows the distribution of the residual regression and the corresponding fitted curve. The identified value of the parameter β_H is summarized in Table A.1.

Table A1
Key parameters required for the physical model.

Parameters	Values
Laser power (P , W)	1140–1500
Scanning-speed (V , mm/s)	10–15
Powder feeding rate (F , g/min)	17.1–20.7
Substrate Melting Point (T_m , K)	1753
Thermal conductivity (k , W/m·K)	22.3(293 K)
Density of Metal Powder (ρ , Kg/m ³)	7800(293 K)
Laser head height (h_0 , mm)	15
Radius of the laser spot (r_b , mm)	1.5
Powder utilization rate (η_p)	0.7[50]
Calibration coefficient of H (β_H)	2.1
Substrate absorption rate (β_w)	0.65[26]
Specific heat capacity of substrate (C_p , J/kg K)	0.45
Brewster effect coefficient (a_w)	0.0196[26]
Temperature calibration factor (n)	0.9
First-order coefficient of attenuation power (k_1 , g ⁻¹ /min ⁻¹)	5.03E-3
Second-order coefficient of attenuation power (k_2 , g ⁻² /min ⁻²)	4.97E-6
First-order attenuation coefficient for depth penetration calibration (k_3 , g ⁻¹ /min ⁻¹)	3.55E-2
Second-order attenuation coefficient for depth penetration calibration (k_4 , g ⁻² /min ⁻²)	1.52E-3

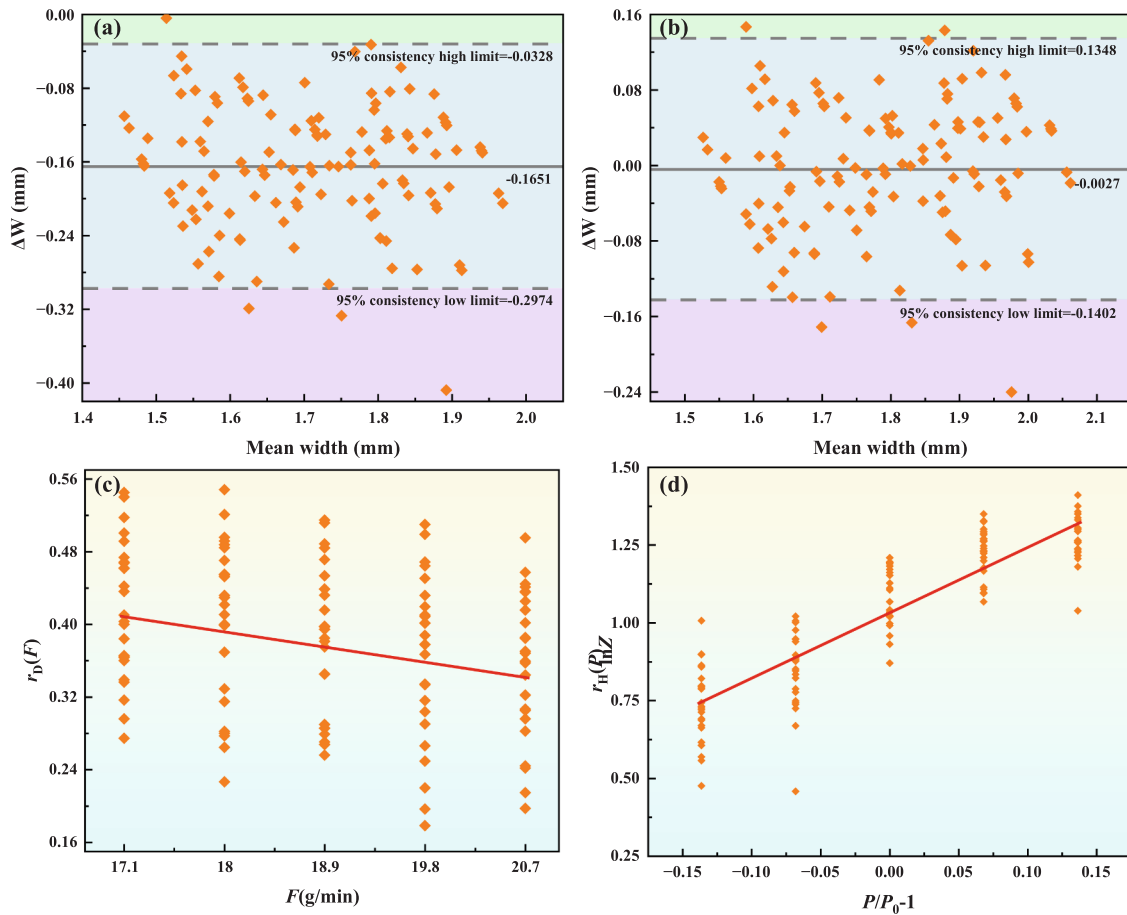


Fig. A1. Schematic illustration of the derivation process for selected key parameters. (a) Melt pool width residual distribution before optimization at $n = 0.8$; (b) melt pool width residual distribution after optimization at $n = 0.9$; (c) regression distribution and fitting for the melt pool depth correction factor; (d) regression distribution and fitting for the melt pool height correction factor.

Data availability

Data will be made available on request.

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